Classification fine-tuning collapses differentiation gradients in single-cell foundation models: a cross-architecture spectrum

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Abstract

Supervised fine-tuning of single-cell foundation models is routine practice, yet whether it preserves the continuous differentiation geometry of pretrained representations has not been quantified. Using the human corneal epithelium, a four-stage differentiation continuum with known ordinal structure, we show that classification fine-tuning collapses the differentiation gradient in a cross-architecture spectrum. Collapse occurrence is corpus-determined: on same-corpus tissue, Geneformer-V2-104M already degrades the gradient under its official one-epoch budget (Spearman ρ 0.806→0.096), and scGPT-human under its official 15-epoch budget shows moderate loss (0.789→0.44, best-val), whereas out-of-corpus and cross-tissue effects remain mild or reversed (0.707→0.641; 0.24→0.46). Collapse determinism is architecture-determined: Geneformer collapses deterministically to noise (three-epoch ρ −0.008), while scGPT-human enters a seed-dependent bistable state (final-epoch ρ −0.09 to 0.83). An ordinal head rescues the gradient in both architectures (ρ 0.833 and 0.819). We release a contamination-aware benchmark pairing same-corpus and out-of-corpus tests.

Keywords: single-cell foundation models, classification fine-tuning, differentiation gradient, ordinal regression, contamination-aware benchmark, corneal epithelium

1. Introduction

The corneal epithelium is maintained by a tightly regulated differentiation continuum from quiescent limbal stem cells (LSCs) at the limbal palisades of Vogt, through transit-amplifying cells (TACs), to committed basal epithelial cells, and ultimately to terminally differentiated squamous cells forming the protective ocular surface barrier [1]. This ordered progression is functionally required: disruption at any stage (LSC deficiency, aberrant amplification, or premature terminal differentiation) compromises corneal transparency and may lead to irreversible vision loss [2].

Single-cell RNA sequencing (scRNA-seq) has resolved this continuum at single-cell resolution, with marker genes for LSCs (e.g., *TCF4*, *POSTN*, *ABCG2*, and *TSPAN7* [3]; *TP63* and *KRT15* [4, 5]), TACs (e.g., *UBE2C*, *CDC20*, and *CENPF* [6]; *MKI67* and *BIRC5* [3]), and differentiated populations (e.g., *KRT12* and *KRT3* [4, 7, 8]) identified across independent

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studies and aggregated in community atlases such as the human ocular surface atlas in CELLxGENE Discover [9]. Yet these efforts produce study-specific embeddings that are not directly comparable across datasets, and no unified reference model maps individual cells to a shared, ordered differentiation-state space. The closest approach, a corneal cell-state reference model with support-vector-machine classification (cPredictor) [10], provides a supervised classifier rather than an evaluation of foundation-model representations or their geometric structure.

Foundation models pretrained on massive single-cell corpora offer a potential solution. Geneformer, a 12-layer Transformer pretrained on approximately 104 million cells from 2,903 datasets [11, 12], and related architectures such as scGPT [13] and scBERT [14], learn contextual gene–gene representations that transfer across tissues. Systematic evaluations have tempered expectations: a unified benchmark of thirteen single-cell foundation models found classical PCA remains competitive in several settings, with pretrained embeddings advantageous mainly in low-label and transfer-learning scenarios [15]; biology-driven reanalyses questioned the pre-train-then-fine-tune paradigm [16]; and large-scale evaluations found that, for cell-type annotation, frozen or zero-shot usage generally underperforms fine-tuning, and freezing pretrained layers was not recommended [17]. Conversely, foundation models designed for zero-shot use can organize developmental lineages and differentiation hierarchies without task-specific training [18], indicating that pretrained representations can encode continuous differentiation structure. Benchmark-driven evaluation has become an established genre in this journal, spanning single-cell foundation models [19, 20] and clinical question answering [21], and representation-quality assessment is emerging as a distinct concern in embedding evaluation [22]. Whether general-purpose representations adequately capture the ordered structure of a specific differentiation hierarchy, such as the corneal continuum, remains open; conversely, task-specific fine-tuning may improve discriminative accuracy while inadvertently collapsing the continuous geometric relationships among adjacent differentiation states that are biologically essential.

The fine-tuning literature has focused on whether adaptation improves classification accuracy and whether it causes forgetting of learned cell-type knowledge; full fine-tuning induces catastrophic forgetting at the label level, motivating parameter-efficient remedies [23]; a complementary line reports bounded fine-tuning gains and proposes marker-gene-guided post-pretraining instead [24]. With respect to geometry, a zero-shot benchmark reported geometric collapse of foundation-model embeddings during cellular dynamics reconstruction, with HVG/PCA outperforming them for trajectory recovery [25], and EMP-FM showed that naive fine-tuning often fails to resolve intermediate cell states under a dual classification–pseudotime objective [26]. Layer-resolved dissections of single-cell foundation models have begun to probe where biological information resides within the network [27, 28], and fine-tuned representations are already deployed in production annotation protocols without quantifying whether adaptation preserves ordinal structure [29]; a related cautionary literature documents cases in which well-performing models fail in settings that matter [30, 31]. What remains absent is a quantified measurement, under controlled fine-tuning conditions, of whether supervised adaptation preserves or destroys the continuous differentiation-state gradient that orders a hierarchy, which we term the *continuity problem*. The corneal epithelium is an ideal test system: its four ordered stages have well-established markers, so the expected ordinal structure is known *a priori* and gradient preservation can be measured directly in embedding

space. We evaluate a family of candidate gradient-preserving fine-tuning recipes under an identical locked protocol, including a continuity-regularized objective (CAFT), a cumulative ordinal head (CORN), an axis-anchored objective (AA-FT), a soft-Spearman rank surrogate, and a flow-consistency objective; whether any of them preserves ordered geometry at matched classification accuracy is a hypothesis to be tested on externally held-out, source-disjoint data, not an assumption.

Here we quantify the effect of classification fine-tuning on differentiation-state geometry in the normal human corneal epithelium. Using a donor-disjoint fair-split protocol on a 10-study corneal and ocular surface atlas (99,606 training cells from 49 donors; 20,394 test cells from 25 donors; zero donor overlap), we compare two reference arms (HVG/PCA with a linear classifier; frozen Geneformer-V2-104M with a linear probe) with adapted configurations sharing identical preprocessing, data splits, classification head, and training budget: naive cross-entropy fine-tuning, a continuity-regularized objective (CAFT), an ordinal head (CORN, with and without the continuity term), and three further gradient-preserving objectives (AA-FT, soft-Spearman, flow), evaluated on a same-corpus donor-disjoint holdout (GSE153515; 17,587 cells, 2 donors), a truly out-of-corpus external test (GSE299152; 21,374 epithelial cells, 3 donors), and a cross-tissue epidermal test set. Our contributions are fivefold: (1) a quantified, three-domain measurement of fine-tuning-induced differentiation-gradient collapse: catastrophic on same-corpus data (the *topology-preserving ordinal-axis collapse*, TP-OAC, layer-resolved across all thirteen hidden states), mild on truly out-of-corpus data, and reversed on cross-tissue data; (2) an ordinal head reparameterization (CORN) as the only recipe preserving the gradient at matched classification accuracy, and only within the pretraining corpus; (3) a systematic evaluation of five alternative gradient-preserving objectives (three honest negatives, one partial, one unstable), in the spirit of recent critical-evaluation studies in this journal [31, 32]; and (4) a contamination-aware scFM benchmark for the normal human corneal epithelium, pairing a same-corpus donor-disjoint holdout with a truly out-of-corpus external test under a locked multi-seed protocol with BH-FDR control and permutation nulls; and (5) a cross-architecture same-corpus arm (scGPT-human on van Zyl) together with official-budget controls, showing that collapse occurrence is corpus-determined and collapse determinism is architecture-determined (Geneformer: deterministic; scGPT-human: seed-dependent bistable). These measurements define a collapse mode we term *topology-preserving ordinal-axis collapse* (TP-OAC), in which the ordinal differentiation axis is erased while discrete layer topology survives; it is distinct from the ordinal geometry that ordinal supervision induces [33], from pretrained layer profiles characterized without adaptation [34], and from the trajectory-fidelity trade-offs that trajectory benchmarks quantify across embedding methods [35]. Unlike prior evaluations of embedding quality, which focused on visualization stability [22] or zero-shot geometry [25], our measurements quantify how supervised adaptation itself alters ordinal differentiation geometry under controlled fine-tuning conditions. Our scope is restricted to normal human corneal epithelium; evaluation of disease states, including dry eye disease, is deferred to future work.

2. Results

2.1. Fine-tuning collapses the differentiation gradient while preserving layer topology

Classification fine-tuning collapsed the differentiation gradient of the corneal epithelium while leaving layer topology intact. Two non-adapted references (HVG/PCA with a linear classifier; frozen Geneformer-V2-104M with a linear probe) were compared against end-to-end cross-entropy fine-tuning and a continuity-regularized variant (CAFT), all sharing identical preprocessing, data splits, classification head, and training budget, on a same-corpus, donor-disjoint holdout (GSE153515; Figure 1; three independent seeds per adapted arm; per-seed values in Table S1).

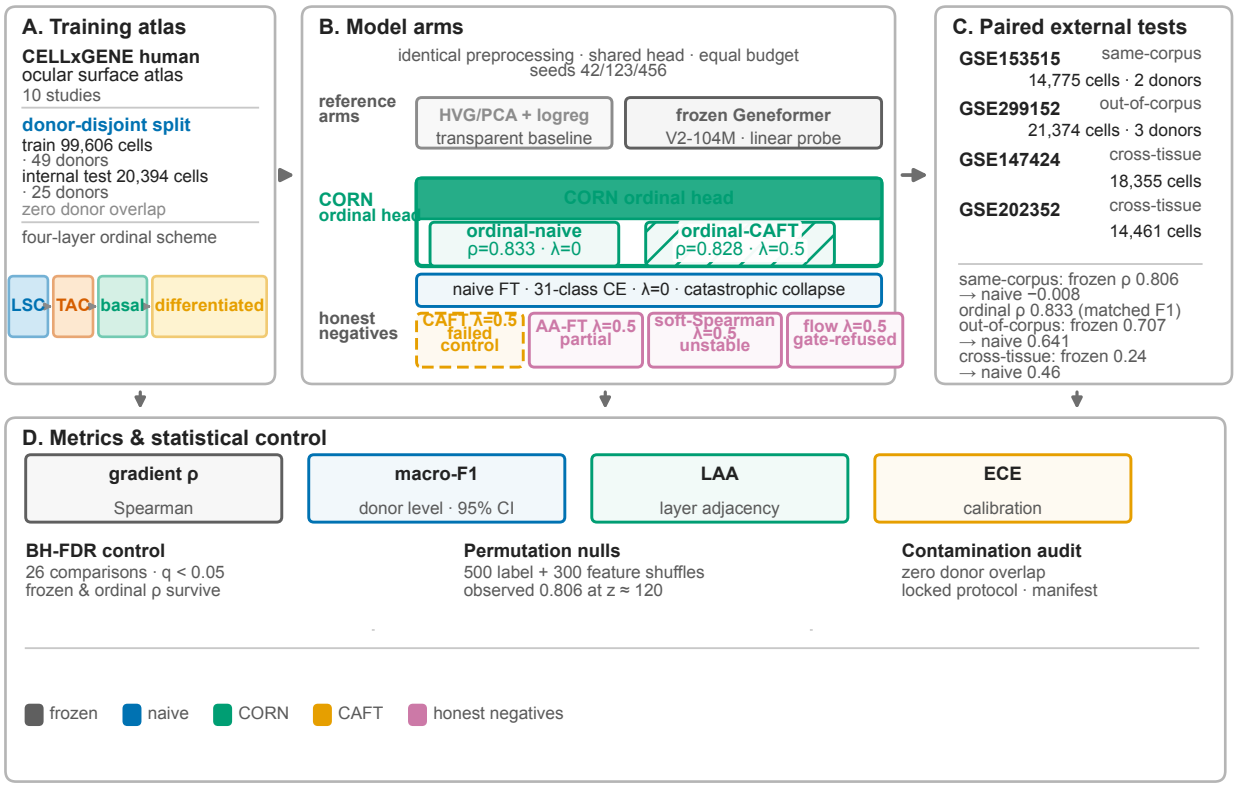


Figure 1: Study pipeline. Donor-disjoint training atlas (CELLxGENE), locked evaluation protocol, softmax-head and ordinal-head fine-tuning arms, and paired same-corpus (GSE153515) and post-cutoff out-of-corpus (GSE299152) external tests.

The two non-adapted reference arms preserved strong differentiation-gradient rank correlation on the same-corpus holdout: Spearman ρ was 0.8206 for HVG/PCA and 0.8064 for frozen Geneformer (Table 2). By contrast, naive supervised fine-tuning collapsed this ordinal signal to near zero: per-seed ρ ranged from -0.042 to $+0.054$ across seeds (three-seed mean -0.0082), with bootstrap 95% confidence intervals spanning zero in every run (Figure 2, Table 2). The sign of ρ was inconsistent across seeds: the residual ordinal correlation after fine-tuning was indistinguishable from noise. A randomized-head ablation – replacing the trained classifier with an untrained head before re-embedding – leaves the hidden states

unchanged, ruling out that the collapse is a readout artifact of the trained classifier (Methods); the erasure resides in the fine-tuned body weights. The biological consequence of this collapse is quantified below (Figure 7).

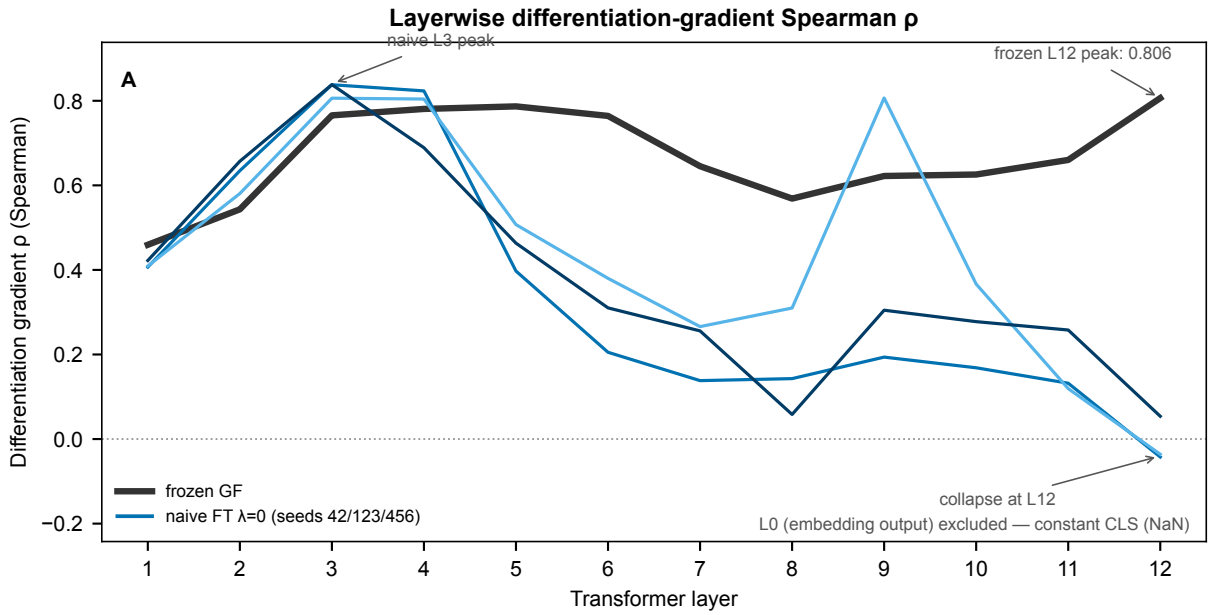


Figure 2: Layer-resolved analysis localizes the collapse to upper transformer layers. Layer-resolved differentiation-gradient Spearman ρ across hidden states L1–L12 for the Geneformer-based configurations (n = 14,775 external-test epithelial cells; three seeds per adapted arm; L0 excluded: constant [CLS]). Color key: thick dark gray line, frozen Geneformer; blue family, naive fine-tuning (seeds 42/123/456). The collapse is head-proximal: early layers retain the gradient, while L5–L12 decay. See also Figure S7 and Table S5.

Despite this gradient collapse, discrete layer topology was preserved. Layer adjacency accuracy (LAA) remained high in all four arms: 0.9771 (HVG/PCA), 0.9665 (frozen Geneformer), 0.9768 (naive, three-seed mean), and 0.9773 (CAFT, three-seed mean). Classification fine-tuning therefore disrupted the continuous ordering of the differentiation continuum while leaving local layer-adjacency structure essentially intact: continuous gradient geometry and discrete layer topology are dissociable under task adaptation.

The continuity-regularized arm must be read as a failed control. A post hoc audit of the archived training code established that the continuity term never executed in any run, because of an implementation defect in the training hook, so those checkpoints trained pure cross-entropy, identical in objective to the naive arm (Methods, Provenance and contamination audit). The +0.210 separation in final-layer ρ therefore reflects run-to-run nondeterminism rather than a causal effect of the regularization, and classification performance was likewise indistinguishable: macro-F1 was 0.6451 (95% CI 0.6164–0.6738) for the continuity-regularized group versus 0.6492 (95% CI 0.6233–0.6751) for naive fine-tuning ($\Delta = -0.0041$; overlapping intervals). Throughout, we report this arm as an honest negative implementation result rather than a promoted safeguard. Fine-tuning nonetheless achieved the highest classification accuracy while collapsing the gradient to noise: differentiation geometry and classification performance are separable.

Layer-resolved analysis localized the collapse to upper transformer layers. To determine

whether fine-tuning erases the differentiation-gradient signal globally or damages it in a layer-localized manner, we re-measured ρ at all thirteen hidden states (embedding output L0 and layers L1–L12; L0 excluded because the position-0 [CLS] representation is constant across cells) for all seven Geneformer-based configurations on the 14,775 external-test epithelial cells; the layer-resolved pipeline reproduced each model's single-value measurement exactly (Methods, Layer-resolved gradient measurement).

The collapse was concentrated in head-proximal layers, with the gradient intact in early layers (Figure 2). First, early-layer representations preserved the gradient in every arm: L1–L4 ρ lay between 0.41 and 0.84, and the fine-tuned models reached their maximum at L3 (0.81–0.84 across seeds), matching or exceeding the frozen model at the same depth (L3, 0.7656). Second, the collapse observed at the final layer of the naive models was concentrated in the upper half of the network: from L5 (0.3973, seed 42) ρ decayed toward zero or negative values at L12, whereas the frozen model's gradient signal rose to its network-wide maximum of 0.8064 precisely at L12. Third, the continuity-regularized checkpoints ended at L12 ρ of 0.19–0.23, above the naive group's near-zero values but far below the frozen model's 0.8064; as the continuity term never executed (Methods), this difference is not attributable to the regularization objective.

Bootstrap confidence intervals at L12 (n = 1,000, cell-level resampling) quantified the separation: frozen $\rho = 0.8064$ [0.7918, 0.8164]; all naive seeds spanned zero; and the continuity-regularized checkpoints were positive (0.19–0.23) and, in this sample, separated from both groups. Because the continuity term never executed in that arm, the separation is a property of these particular cross-entropy runs, not evidence of a regularization effect. Gradient collapse under classification fine-tuning is therefore a head-proximal overwrite: the signal persists in intermediate layers and is selectively disrupted in the layers nearest the classification head.

To test whether this overwrite reflects the training process rather than an artifact of the trained classification head, we replaced the trained head of the naive seed-42 checkpoint with a freshly initialized, untrained random head: the layerwise differentiation-gradient ρ profile remained bit-identical at all thirteen layers (L12 $\rho = -0.0424$), while the same treatment on the frozen body preserved the frozen profile exactly (L12 $\rho = 0.8064$). The L5–L12 collapse is therefore intrinsic to the fine-tuned body weights, a training-induced rewrite of head-proximal layers, and not a readout artifact of the trained head (Methods, Layer-resolved gradient measurement).

Trajectory inference confirmed the geometric dissociation at the level of inferred cell ordering (Supplementary Figure S12). Diffusion pseudotime on the same external cells, rooted at the LSC class, correlated strongly with the reference layer-ordering pseudotime for the frozen model ($\rho = 0.850$, 95% CI 0.842–0.858) but weakly for naive fine-tuning ($\rho = 0.323$–0.348), with CAFT intermediate ($\rho = 0.372$–0.428). The loss of ordinal information under naive fine-tuning is thus observable not only in our gradient metric but in a widely used trajectory-reconstruction procedure.

2.2. The collapse is domain-specific: a three-domain boundary map

Having measured the collapse on a same-corpus holdout, we asked whether it reflects the evaluation domain rather than fine-tuning itself, and delineated its boundary across three domains of increasing distance from the pre-training corpus. To separate the effect of fine-tuning from the effect of pre-training exposure, we paired two external test sets

with opposite corpus status and added a cross-tissue control: GSE153515 (normal limbal epithelium), whose primary publication DOI resolves to a row in the official Genecorpus-104M source table (53,290 cells, organ = eye) and thus lies within the pre-training corpus (least conservative); GSE299152, a single-cell map of the ocular surface glycocalyx [36], deposited after corpus assembly and absent from the table (post-cutoff, most conservative); and human epidermis (GSE147424 [37]) as a cross-tissue control. Construction and annotation of these sets follow the locked protocol in Methods (Supplementary Figure S13).

On the same-corpus holdout, fine-tuning drove performance while collapsing the gradient. On GSE153515, the HVG/PCA reference arm achieved donor-level macro-F1 of 0.6192 (95% CI 0.5747–0.6637), outperforming frozen Geneformer (0.4421; 95% CI 0.3749–0.5093), the opposite of the internal training-split ranking (0.8834 versus 0.7787). Frozen Geneformer showed poor calibration on external data (ECE 0.3010) and basal-class collapse (recall 0.216), whereas HVG/PCA remained well calibrated (ECE 0.0145) with strong gradient correlation. Both fine-tuned arms (naive 0.6492; CAFT 0.6451) outperformed both baselines. These results suggest that fine-tuning, not pretraining alone, drives cross-dataset generalization (Figure 3); the magnitude of these scores sits in the range expected for donor-disjoint validation, in which external macro-F1 falls well below internal-split numbers across the single-cell foundation-model literature [15, 16]. On this same-corpus holdout, the reference arms preserved the differentiation gradient (frozen $\rho = 0.8064$), whereas naive fine-tuning collapsed it to -0.008 with preserved layer topology (TP-OAC, above). The two evaluation dimensions are thus orthogonal at the level of individual arms: frozen Geneformer is best at geometry ($\rho = 0.8064$) but poorest at classification (macro-F1 0.4421), whereas the fine-tuned arms are best at classification (0.6492) but collapse the geometry; classification accuracy and gradient fidelity must be audited separately.

Both adapted arms generalized across the two donors (donor_1 > donor_2 in all six runs; Table 2). Because the TAC class is extremely rare (171/14,775 cells; 1.2%) and received zero true-positive predictions in every adapted run, we performed a sensitivity analysis excluding TAC (Supplementary Figure S10). Three-class macro-F1 rose by +0.21–0.22 in all adapted arms (naive 0.8599–0.8763; CAFT 0.8479–0.8786) and by +0.15 for frozen Geneformer (0.5921); the three-class confidence intervals did not overlap the four-class intervals, confirming that TAC rarity systematically depresses the primary endpoint. The naive-versus-CAFT comparison remained non-significant under the three-class definition, and the arm ranking was unchanged.

Out-of-corpus and cross-tissue domains bounded the collapse (Table 1, Figure 4). On the truly out-of-corpus test (GSE299152), classification fine-tuning collapsed the gradient only mildly: frozen $\rho = 0.707$ fell to 0.641 under naive fine-tuning (three-seed mean 0.641 ± 0.030), a partial rather than catastrophic loss, with a stable sign across seeds. The ordinal recipe did not transfer: ordinal-head ρ ranged from 0.35 (ordinal-naive; mean 0.350 ± 0.099) to 0.54 (ordinal-CAFT; mean 0.539 ± 0.007), below the naive arm's 0.641, and macro-F1 fell correspondingly (0.29–0.30 versus 0.452). The CORN head therefore preserves the gradient only within the pre-training corpus, where the ordinal axis is already encoded; out-of-corpus, its threshold predictions are poorly calibrated (ECE 0.56–0.61) and underperform cross-entropy.

On the cross-tissue domain (human epidermis [37], GSE147424; four ordered keratinocyte layers), the direction of the effect reversed: fine-tuning *improved* the gradient, from frozen $\rho =$

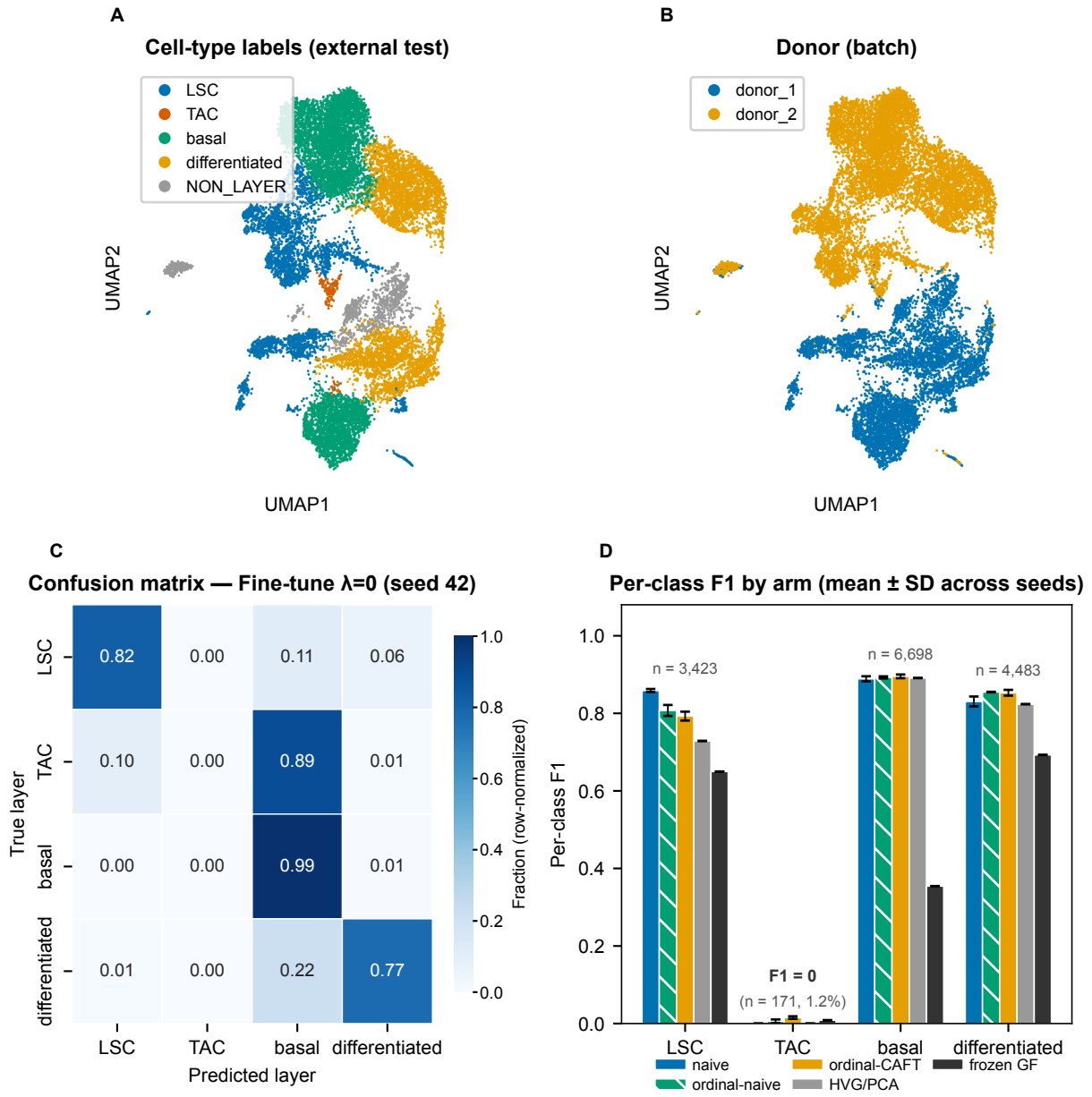


Figure 3: Same-corpus holdout: fine-tuning drives performance and collapses the gradient. External evaluation of adapted models on GSE153515 (n = 14,775 epithelial cells, two donors). (A) Cell-type UMAP. (B) Donor UMAP. (C) Confusion matrix. (D) Per-class F1 across five model arms (adapted arms three-seed means; reference arms single runs); the TAC column shows F1 = 0 (n = 171; 1.2%; three-class sensitivity in Figure S10). Color key (D): blue, naive; blue-green, ordinal-naive (hatched); orange, ordinal-CAFT; gray, HVG/PCA; dark gray, frozen. See also Figure S6.

0.24 to $\rho = 0.46$ under naive fine-tuning (ordinal head 0.50). Notably, this inversion cannot be attributed to corpus absence: the DOI-level audit (Methods, Provenance and contamination audit) resolves GSE147424 to a row in the Genecorpus-104M source table (organ = skin, 45,200 cells), so the pretrained model did see epidermis, yet encoded its differentiation axis only weakly ($\rho = 0.24$ versus 0.81 for the cornea). The discriminating factor is therefore the strength of the tissue-specific differentiation axis that pretraining encodes, not corpus membership per se: classification supervision overwrites an axis it can already read (cornea) and builds one where pretraining left it weak (epidermis). Together, the three domains delineate the boundary of TP-OAC: catastrophic on the in-corpus tissue with a strongly encoded axis, mild on post-cutoff out-of-corpus data, and absent (inverted) on the in-corpus tissue whose axis pretraining encodes weakly.

Table 1: Differentiation-gradient Spearman ρ (frozen, naive cross-entropy fine-tuning, and ordinal head) across three evaluation domains. Same-corpus adapted-arm values are three-seed means with per-seed ranges in parentheses; out-of-corpus adapted-arm values are three-seed mean ± SD, with the ordinal range spanning the ordinal-naive and ordinal-CAFT arm means; frozen and cross-tissue values are single runs. Statistical control (BH-FDR and permutation nulls) applies to the same-corpus family.

Domain	Frozen	CE-naive	Ordinal (COI
Same-corpus (GSE153515)	0.806	−0.008 (−0.042 to +0.054)	+0.833 (0.830–0.8
Out-of-corpus (GSE299152)	0.707	0.641 ± 0.030	0.350 ± 0.099 to 0.539 ± 0.
Cross-tissue epidermis (GSE147424)	0.240	0.460	0.
Cross-tissue epidermis (GSE202352)	0.520	0.710	0.

All primary same-corpus claims survived statistical control: a single BH-FDR family of 26 comparisons (q threshold 0.05) and permutation nulls (Methods, Statistical analysis plan). Frozen and ordinal ρ were significantly positive (both $q = 0.0086$), the naive collapse was not ($q = 1.0$), and the frozen observation sat more than 120 standard deviations from the label-shuffle null.

2.3. Alternative gradient-preserving recipes: honest negatives, a partial fix, and an unstable one

Is the collapse avoidable? Beyond the ordinal head, we evaluated five alternative gradient-preserving recipes under the identical locked protocol (Methods, Model architectures): three failed outright, one preserved the gradient only partially, and one was unstable across seeds (per-arm three-seed values in Table S9). CAFT, whose continuity objective never executed because of an implementation defect, is an honest negative: its higher final-layer ρ (+0.210 versus −0.008) was run-to-run nondeterminism rather than restoration (Methods, Provenance and contamination audit). A correctly wired continuity term (L_{cont}, tested inside the ordinal-CAFT arm) conferred no measurable benefit beyond the ordinal head ($\Delta\rho = 0.005$), echoing the auxiliary-signal suppression observed in frozen models [38]. The axis-anchored objective preserved the gradient only partially (three-seed mean 0.208), and the rank-based objective was unstable across seeds (mean 0.095, SD 0.172); flow was refused extension by its stability gate. Single-axis recipes cannot substitute for the ordinal head: a fixed axis captures at most the centroid-trajectory direction, not the full layer geometry that the metric derives per checkpoint (Methods, Axis definition revision).

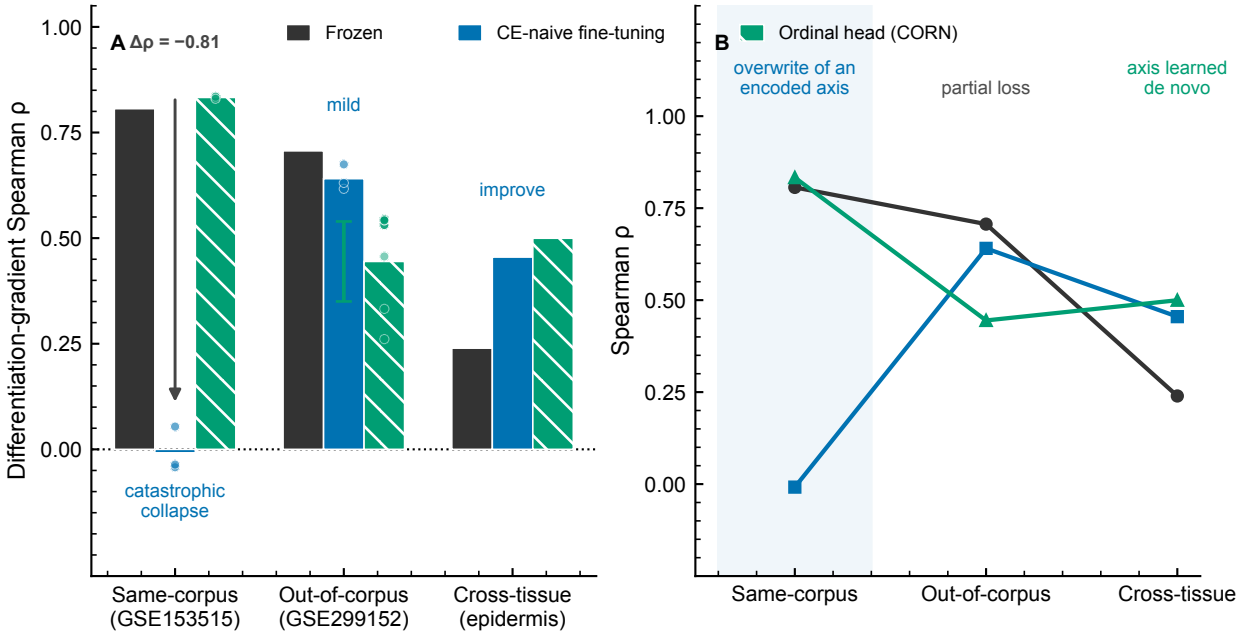


Figure 4: The collapse is domain-specific: a three-domain boundary map. (A) Spearman ρ by evaluation domain and arm (frozen, CE-naive, ordinal head); same-corpus adapted values are three-seed means, out-of-corpus ordinal range as mean ± SD (n = 14,775 same-corpus and 21,374 out-of-corpus epithelial cells; counts per domain in Methods). (B) Frozen-to-naive trajectories across domains: catastrophic overwrite within-corpus, partial loss out-of-corpus, and de novo axis learning cross-tissue. Color key: dark gray, frozen; blue, CE-naive; green, ordinal head (hatched). See also Table S5.

2.4. An ordinal head parameterization preserves the differentiation gradient

The honest negative leaves open whether any fine-tuning recipe can preserve the differentiation gradient at matched accuracy. We therefore tested a constructive alternative, reparameterizing the classification head along the known ordinal axis instead of treating the four layers as unordered classes. Following the cumulative ordinal regression (CORN) formulation [39], the head emits $K-1=3$ binary threshold logits, where threshold k predicts whether the cell has reached ordinal stage k or beyond; the loss is the sum of the three binary cross-entropies (masked to epithelial cells), and inference recovers the predicted layer as the number of thresholds with sigmoid activation ≥ 0.5. The CORN head makes no equidistance assumption on the thresholds and yields class probabilities directly for calibration ($P(y=k)=p_k-p_{k+1}$).

Two ordinal arms were trained under the identical locked protocol and budget: ordinal-naive ($\lambda=0$; ordinal loss only) and ordinal-CAFT ($\lambda=0.5$; ordinal loss plus the CAFT continuity term, here wired through a registered Trainer callback and therefore genuinely executed, in contrast to the archived CAFT arm). Both preserved the differentiation gradient at classification accuracy matched to the softmax arms (Table 2): three-seed mean ρ was 0.8330 (per-seed 0.8299–0.8351) for ordinal-naive and 0.8277 (0.8242–0.8321) for ordinal-CAFT, versus -0.008 for naive softmax fine-tuning, while donor-level macro-F1 was 0.6490 (95% CI 0.6100–0.6879) and 0.6479 (95% CI 0.6099–0.6860), statistically indistinguishable from the naive arm's 0.6492 (bootstrap 95% CI overlap). Layer adjacency accuracy remained $\approx$0.97 in both ordinal arms.

The near-equality of the two ordinal arms ($\Delta\rho=0.005$) indicates that the gradient preservation reflects the ordinal head parameterization itself rather than the continuity term: encoding the a priori layer order as ordered thresholds is sufficient, and the correctly wired centroid-adjacency regularization adds no measurable benefit on this axis. This is the constructive counterpart to the CAFT honest negative: a head reparameterization that respects the known ordinal structure preserves gradient geometry at matched classification performance, whereas the unordered softmax head erases it (Figure 5). This constructive result is confined to the same-corpus domain: on the truly out-of-corpus test the ordinal head preserved less gradient than cross-entropy fine-tuning (Table 1), indicating that the recipe operates by conserving an ordinal axis that pre-training has already encoded. The geometric preservation also comes at a calibration cost: on the same-corpus holdout, ordinal-naive ECE (0.1251) and ordinal-CAFT ECE (0.0609) were not better than the naive arm's 0.0575, and out-of-corpus the ordinal thresholds were markedly worse calibrated (ECE 0.56–0.61). The CORN head therefore trades calibration for geometry: it is a geometry-faithful readout whose probability estimates should be interpreted cautiously, particularly outside the pre-training corpus. The learned decision boundaries are themselves ordered, not merely accurate (Figure S11): the three threshold logits shift monotonically across differentiation stages, per-stage predicted probabilities align with the a priori order, and the per-cell cumulative monotonicity $P(y\geq 1)\geq P(y\geq 2)\geq P(y\geq 3)$ holds in 0.72 of cells versus 0.18 expected from an unordered head (three-seed mean 0.79; Supplementary Figure S11). Although the TAC class received no true-positive predictions in any adapted arm (F1 = 0), the ordinal head does not discard it: CORN threshold probabilities place TAC cells between LSC and basal (median predicted basal probability 0.975; Figure S11B), the biologically expected ordinal position, so the head's response to class rarity preserves rather than erases ordinal structure.

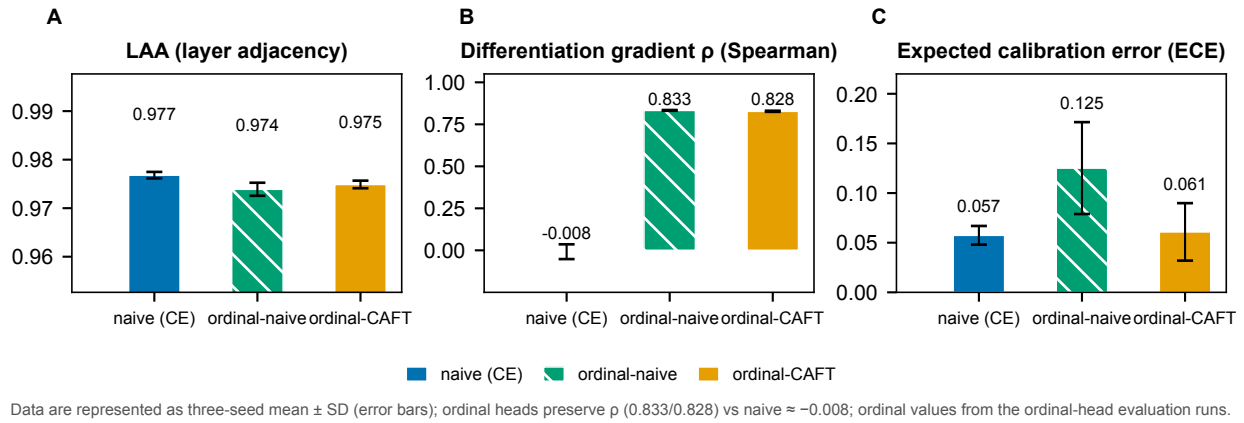


Figure 5: An ordinal head parameterization preserves the differentiation gradient. Continuity metrics under task adaptation. (A) Layer adjacency accuracy (LAA). (B) Differentiation-gradient Spearman ρ. (C) Expected calibration error (ECE). Bars denote the three adapted arms: naive (blue), ordinal-naive (blue-green, hatched), and ordinal-CAFT (orange) (three-seed mean ± SD; n = 14,775 cells per arm). See also Figure S3.

Table 2: External test performance by model arm (GSE153515, four classes; n = 14,775 epithelial cells from two donors). Adapted-arm values are three-seed means with donor-level bootstrap 95% CIs; reference arms are single runs (seed 42). LAA, layer adjacency accuracy; ECE, expected calibration error.

Model arm	Macro-F1	95% CI	LAA	ρ	ECE
HVG/PCA + linear	0.6192	[0.5747, 0.6637]	0.9771	0.8206	0.0145
Frozen Geneformer	0.4421	[0.3749, 0.5093]	0.9665	0.8064	0.3010
Naive FT ($\lambda = 0$)	0.6492	[0.6233, 0.6751]	0.9768	−0.0082	0.0575
CAFT ($\lambda = 0.5$)	0.6451	[0.6164, 0.6738]	0.9773	+0.2102	0.0600
Ordinal-naive ($\lambda = 0$)	0.6490	[0.6100, 0.6879]	0.9739	+0.8330	0.1251
Ordinal-CAFT ($\lambda = 0.5$)	0.6479	[0.6099, 0.6860]	0.9749	+0.8277	0.0609

2.5. The collapse is a cross-architecture spectrum: corpus determines occurrence, architecture determines determinism

To test whether the collapse measured in Geneformer-V2-104M reflects the architecture or the phenomenon itself, we repeated the same-corpus fine-tuning protocol in scGPT-human, a second single-cell foundation model, on its only in-corpus corneal tissue. scGPT-human was pretrained on the CELLxGENE census 2023-05-15 release [13], whose single corneal dataset is the van Zyl et al. limbal and central cornea atlas [40]. We therefore built a same-corpus van Zyl arm (62,640 epithelial cells after the locked cluster-0 exclusion of 13,584 ciliary-muscle cells; three-tier axis LSC→basal→differentiated, the transit-amplifying class being unfittable because all six cycling markers were removed by the locked gene filter), with the largest-epithelial donor (Hu0216; 20,158 cells) held out as the locked test donor (Methods).

Both models passed the pre-registered frozen-axis gate (G1: $\rho \geq 0.5$ and label-shuffle $z \geq 3$): scGPT-human $\rho = 0.8047$ ($z = 229.9$) and Geneformer $\rho = 0.8292$ ($z = 153.6$) on the same van Zyl three-tier axis, and the revised training-adequacy gate (G0: $\mathrm{F1}_{\mathrm{ce}} \geq 0.55$ and $\geq \mathrm{F1}_{\mathrm{frz}} - 0.10$) passed in all three scGPT seeds. Classification fine-tuning then degraded the gradient in both architectures, but with architecture-dependent determinism (Figure 6). Under the official one-epoch Geneformer budget, naive fine-tuning already collapsed the same-corpus gradient: three-seed $\rho = 0.096$ (per-seed 0.201, 0.144, −0.057), versus −0.008 at the three-epoch budget of the main protocol (Table 4); collapse therefore does not require overtraining. scGPT-human, trained to its official 15-epoch budget with per-epoch validation checkpoints, showed moderate average loss but pronounced seed heterogeneity: the best-validation checkpoints (selected by validation macro-F1, the scGPT official convention) reached test-donor $\rho = 0.34$–0.58 (three-seed mean 0.44; macro-F1 0.920–0.942), whereas the final-epoch checkpoints split into a bistable pattern: seed 42 collapsed to $\rho = -0.09$ ($z = -12.2$), seed 123 retained the gradient ($\rho = 0.833$, above the frozen baseline of 0.789), and seed 456 sat near the baseline ($\rho = 0.765$; Figure 6B, Table 3). Validation trajectories were noisy and non-monotonic, with per-seed validation-ρ peaks at epochs 3, 3, and 6 (0.730, 0.852, 0.778) and no deterministic late-epoch collapse (Table 3).

The ordinal recipe rescued the gradient in both architectures: CORN $\rho = 0.833$ in Geneformer (main protocol) and 0.819 in scGPT-human (three-seed mean, range 0.647–0.907). Together with the three-domain boundary map (Table 1), these data delineate the spectrum: collapse occurrence is determined by corpus membership (same-corpus tissue collapses under either architecture at either official budget), whereas collapse determinism is architecture-determined (Geneformer converges deterministically to noise; scGPT-human enters a seed-dependent bistable regime, echoing the seed-grouped convergence-speed heterogeneity documented for scBERT fine-tuning [41]).

2.6. The differentiation axis recovers CNS-published biology

The geometric measurements above leave open whether the ordinal axis carries the biology established for this tissue, or whether the collapse is a rearrangement without biological content. We therefore asked whether each arm's axis recovers two *Cell*-published discoveries: the slow-cycling limbal stem-cell compartment of Cotsarelis et al. [1] (a quiescent limbal stem-cell population that can be stimulated to proliferate, yielding transient-amplifying cells and then terminally differentiated keratinocytes) and the basal-to-suprabasal keratin

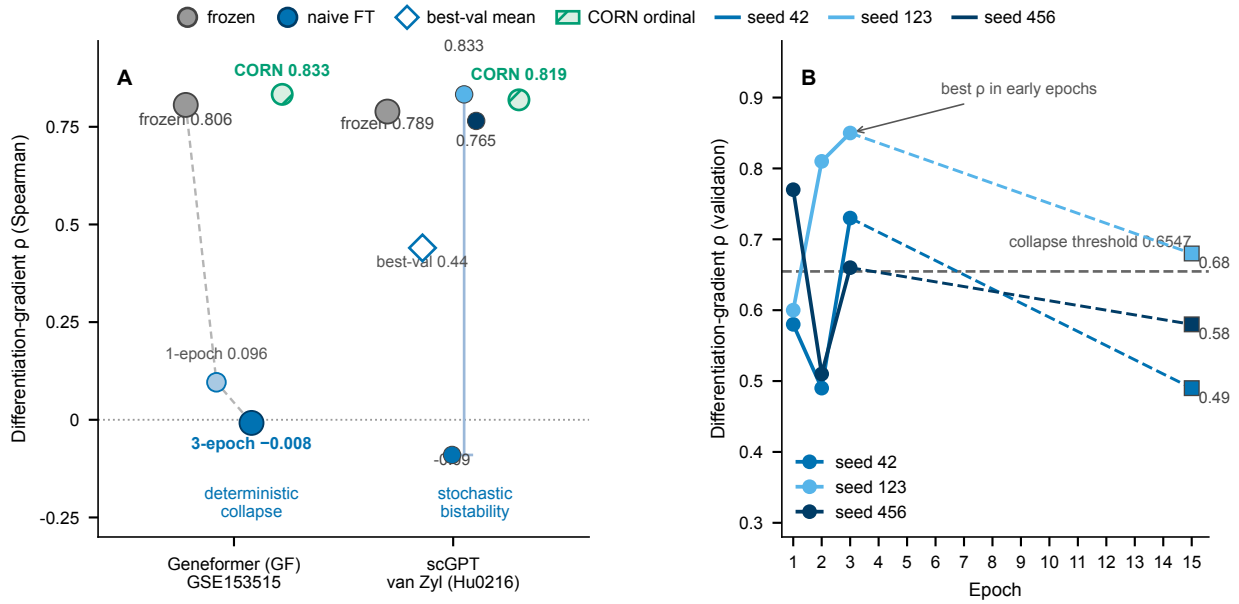


Figure 6: A cross-architecture spectrum of differentiation-gradient collapse. (A) Differentiation-gradient Spearman ρ for Geneformer-V2-104M (same-corpus GSE153515; n = 14,775 epithelial cells) and scGPT-human (same-corpus van Zyl, locked test donor Hu0216; n = 20,158): frozen (dark gray), naive fine-tuning at the official budget (Geneformer one epoch, three-seed mean 0.096; scGPT 15-epoch best-val, three-seed mean 0.44; blue), naive three-epoch fine-tuning (Geneformer three-seed mean −0.008; scGPT final-epoch per-seed −0.09/+0.833/+0.765; light blue), and ordinal head (CORN; ρ 0.833 and 0.819; green). (B) scGPT-human convergence trajectories: per-epoch validation ρ for seeds 42/123/456 (epochs 1–3 solid; epoch-15 endpoints; intermediate epochs dashed), showing non-monotonic, seed-dependent behaviour with the best epochs early; frozen baseline 0.789 (gray line). Adapted-arm values are three-seed means, with per-seed values in Table 3 and label-shuffle z scores in Tables 3 and 4.

Table 3: scGPT-human same-corpus arm (van Zyl, locked test donor Hu0216; n = 20,158 epithelial cells). Convergence-arm checkpoints per seed, evaluated on the test donor: macro-F1 with cell-level bootstrap 95% CI (n = 1,000, seed 42), differentiation-gradient ρ (L2-normed), and label-shuffle z (n = 100); validation-ρ peak and its epoch from the per-epoch trajectories, with best-validation epoch (by validation macro-F1). Frozen baseline (test donor): $\rho = 0.7887$, $z = 99.3$. Single test donor: donor-level statistics degenerate, cell-level bootstrap used throughout (Methods).

Checkpoint	Seed	Macro-F1	95% CI	ρ	z	Val ρ peak (epoch)
Best-val	42	0.9313	[0.9272, 0.9351]	0.3373	36.4	0.730 (3)
Best-val	123	0.9419	[0.9384, 0.9454]	0.5774	66.3	0.852 (3)
Best-val	456	0.9201	[0.9158, 0.9243]	0.4039	41.8	0.778 (6)
Best-val mean		0.9311	[0.9041, 0.9581]	0.4395	48.2	
Final-epoch	42	0.9038	[0.8992, 0.9083]	−0.0908	−12.2	
Final-epoch	123	0.8826	[0.8774, 0.8876]	0.8329	90.2	
Final-epoch	456	0.8773	[0.8722, 0.8826]	0.7647	85.1	
Final-epoch mean		0.8879	[0.8532, 0.9226]	0.5023	54.4	

Table 4: Geneformer official-budget control (same-corpus GSE153515; n = 14,775 epithelial cells). Naive cross-entropy fine-tuning at the official one-epoch budget versus the main three-epoch protocol: per-seed differentiation-gradient ρ (raw) with three-seed means and donor-level macro-F1 per seed. The three-epoch reference is the pre-registered main protocol arm (Table 2).

Budget	ρ per seed (42/123/456)	ρ mean	Macro-F1 per seed
Official one epoch	0.201 / 0.144 / −0.057	0.096	0.6397 / 0.6414 / 0.6357
Main protocol (three epochs)	−0.042 / −0.036 / +0.054	−0.008	0.6542 / 0.6420 / 0.6515

switch of Fuchs and Green [42]. Along the frozen axis the published program is recovered (Figure 7): the stemness program (KRT15, KRT19, TP63) declines monotonically from the stem-cell pole toward differentiation ($\rho = -0.41$; bootstrap 95% CI −0.42 to −0.39), the corneal differentiation program (KRT12, KRT3) rises ($\rho = +0.69$; +0.68 to +0.70), and the cycling program peaks in the transient-amplifying compartment (mean score 3.15 in TAC cells versus < 0.05 in every other class; PCNA and MCM2, both absent from the annotation marker sets, peak in the same compartment). Naive fine-tuning erases the graded structure: the stemness correlation collapses to a three-seed mean of $+0.02 \pm 0.03$ (per-gene: KRT15 $-0.53 \rightarrow -0.06$; KRT19 $-0.41 \rightarrow -0.12$), while the differentiation-pole signal is only partially retained ($+0.69 \rightarrow +0.29 \pm 0.03$), consistent with retained coarse class discrimination (macro-F1 0.65) without the graded axis. The ordinal head, which preserves the gradient geometry, also preserves the biology with negligible seed variance: three-seed stemness ρ was -0.427 ± 0.004 (ordinal-naive) and -0.407 ± 0.012 (ordinal-CAFT), and differentiation ρ $+0.557 \pm 0.005$ and $+0.567 \pm 0.012$, statistically indistinguishable from the frozen reference.

The recovery is not an annotation artifact: the per-gene panel includes markers absent from every locked annotation marker set (NFIB −0.16, CDH3 −0.07, PCNA +0.14 on the frozen axis; NFIB −0.22 on the ordinal axis), and all program correlations exceed the permutation null by tens of standard deviations (stemness $z = -49$; differentiation $z = +82$); TP63 alone is basal-peaked rather than stem-cell-peaked in this 10x dataset and is not axis-separable (+0.13), an honest boundary of single-marker recovery. The same framework transfers to the cross-tissue control, and the reversal is reproduced on a second independent epidermal dataset (GSE202352 [43]; Figure 7C). On GSE147424 the basal keratin program (KRT5, KRT14, COL17A1) declines along the axis in every arm (−0.59 frozen to −0.75 fine-tuned), and the cycling program declines from the proliferative basal compartment (−0.23 fine-tuned), recovering the Fuchs–Green switch in its measurable (basal-off) half; the differentiation-pole keratins are only partially monotone in this dataset (KRT10 +0.18 ordinal; KRT1 bimodal across layers), a limitation we report rather than smooth over. On GSE202352 the frozen axis instead encodes the differentiation-pole keratins (KRT1, KRT10; $\rho \approx +0.83$) rather than the basal pole (+0.11), but every adapted arm builds the basal-off program (−0.65 to −0.79) with the cycling program declining from the proliferative basal compartment (−0.21), and the class-ordinal reversal holds (frozen 0.52 → fine-tuned 0.71–0.76). The frozen baselines thus differ across datasets – the axis encodes different halves of the switch – while adapted arms consistently orient it toward the full basal-to-suprabasal program, and naive fine-tuning erases the graded biology on the in-corpus cornea in lockstep with the geometric collapse. Together with the cross-tissue reversal, these data indicate that the axis is the biology: the

frozen and ordinal embeddings arrange cells along the differentiation programs discovered in *Cell* 1980 and 1989, and naive fine-tuning erases that graded biological structure while leaving layer topology intact.

3. Discussion

The central finding of this study is that differentiation geometry and classification performance are separable properties of single-cell representations. Classification fine-tuning of Geneformer-V2-104M collapsed the differentiation-gradient Spearman ρ from $\approx$0.82 (0.8206 for HVG/PCA; 0.8064 for frozen Geneformer) to -0.008 on the same-corpus holdout, while layer adjacency accuracy remained $\approx$0.97 in all adapted arms and macro-F1 rose to 0.6492. The layer-resolved analysis localizes the loss to a head-proximal overwrite: intermediate layers retain the gradient (L3 ρ 0.81–0.84, matching the frozen model at the same depth), while the readout layers nearest the classification head (L5–L12) are selectively degraded. Pretraining concentrates differentiation structure precisely in the final layer, and classification training rewrites exactly that layer. The practical implication is direct: pipelines that fine-tune foundation models and then reuse the resulting embeddings for analyses that require continuous structure (trajectory inference, pseudotime ordering, state-transition quantification) will lose ordinal information even when classification accuracy is high; where continuous structure matters, intermediate-layer or frozen readouts deserve explicit consideration. Conversely, the ordinal recipe that preserves geometry does so at a calibration price (same-corpus ECE 0.1251 versus 0.0575 for naive; out-of-corpus 0.56–0.61), so the trade-off among gradient fidelity, classification accuracy, and calibration is three-way rather than binary.

Three observations bound this collapse and its mechanism. First, the effect is domain-specific rather than universal: catastrophic on same-corpus data, mild on a truly out-of-corpus test (frozen 0.707 $\rightarrow$ naive 0.641), and reversed on cross-tissue data (frozen 0.24 $\rightarrow$ naive 0.46). The inversion is explained by the strength of the pretrained, tissue-specific differentiation axis rather than by corpus membership: the corpus audit places the epidermal test within Genecorpus-104M (GSE147424 resolves to an organ = skin row; 45,200 cells), yet the frozen corneal axis is strongly organized (ρ = 0.81) while the frozen epidermal axis is weak (ρ = 0.24); classification supervision overwrites an axis it can already read and builds one where pretraining left it weak. Second, the collapse is recipe-specific: of five gradient-preserving objectives evaluated under an identical protocol, only the ordinal head preserved the gradient robustly, and only within-corpus; the correctly wired continuity term conferred no benefit beyond the ordinal head itself, and single-axis recipes captured at most the centroid-trajectory direction. Together with the head-proximal overwrite, these observations indicate that preserving ordinal geometry under adaptation requires encoding the ordinal structure in the supervisory signal itself, at the readout where the collapse occurs.

We frame this pattern as *topology-preserving ordinal-axis collapse* (TP-OAC): supervised adaptation erases the ordinal axis of a pre-existing differentiation continuum while cluster membership and layer topology survive, with the loss graded by depth and concentrated in head-proximal layers. TP-OAC joins the neural-collapse family of phenomena [44–46] yet differs from each of them: neural collapse is the terminal-phase convergence of within-class features, whereas TP-OAC is a cross-class ordinal-axis erasure that leaves discrete topology intact. It is likewise distinct from the ordered geometry that ordinal supervision induces [33],

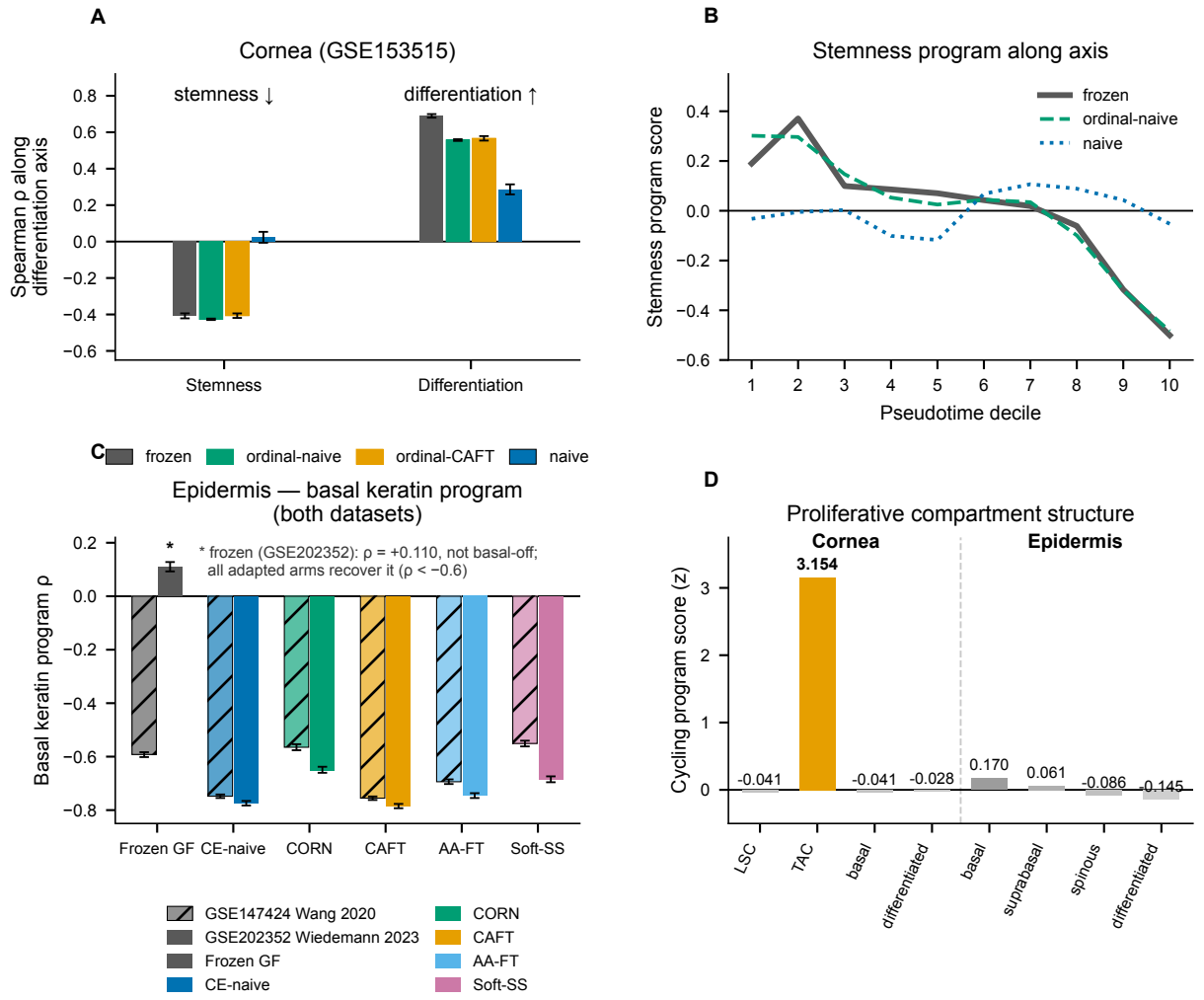


Figure 7: The differentiation axis recovers CNS-published biology. (A) Cornea (GSE153515, n = 14,775): Spearman ρ of stemness and differentiation programs along each arm's axis (frozen with bootstrap 95% CI; adapted arms three-seed mean ± SD). (B) Stemness-program score by pseudotime decile: the graded decline is preserved by frozen and ordinal embeddings and flattened by naive fine-tuning. (C) Epidermis, both datasets (GSE147424, n = 18,355; GSE202352, n = 14,461): basal-keratin program ρ by arm; every adapted arm recovers the basal-off switch, reproducing the cross-tissue reversal. (D) Cycling program by class: the proliferative transient-amplifying compartment (corneal TAC; epidermal basal) sits between the quiescent stem pole and the post-mitotic layers, the slow-cycling structure of Cotsarelis et al. [1]. Color key follows Figure 5. See also Methods, CNS-anchored marker-program validation.

from the intermediate-depth localization of differentiation signal under zero-shot conditions [34], from the generic representational degradation that R3F mitigates [47], and from the label-level catastrophic forgetting documented for parameter-efficient fine-tuning [23]; our randomized-head ablation localizes the effect to the fine-tuned body rather than the readout. Notably, zero-shot evaluations found that Geneformer's embedding space fails to retain discrete cell-type information [48], yet we measured a strong frozen ordinal gradient ($\rho = 0.8064$): ordinal geometry and discrete type identity are dissociable properties of the same representation, and each must be audited separately. The frozen cross-architecture probes extend the architectural scope: scFoundation, UCE, and scGPT-human all separated the four layers but encoded the ordinal axis with markedly different strength ($\rho = 0.75$, 0.58, and 0.22 on GSE153515; Table S8), so ordinal-geometry encoding is architecture-dependent rather than universal. To our knowledge, fine-tuning-induced collapse of an existing ordinal axis, with topology survival, has not previously been measured.

The cross-architecture arm reframes the collapse as a spectrum rather than a Geneformer-specific defect. The same fine-tuning protocol applied to scGPT-human on its in-corpus cornea (van Zyl) reproduced the gradient loss, but not its determinism: across three seeds the final-epoch checkpoints split into a bistable pattern, with one seed collapsing to noise ($\rho = -0.09$) and two retaining near-frozen or above-frozen gradients ($\rho = 0.833$ and 0.765), and the per-epoch validation trajectories were noisy and non-monotonic, with best epochs early. A seed-dependent split between collapse and retention is a basin-of-attraction pattern rather than a deterministic overwrite: the same supervisory objective, differing only in initialization, lands on qualitatively different geometry. This echoes the seed-grouped convergence-speed heterogeneity documented for scBERT fine-tuning, in which low-performing seeds caught up with additional training [41]; in our data, the analogous prescription is checkpoint selection, because the best-validation checkpoints (early epochs) retained the gradient in every seed (test-donor ρ 0.34–0.58) and were the only scGPT configurations whose classification performance met the revised training-adequacy gate. The official-budget control resolves the fairness question raised by the shared three-epoch protocol: Geneformer collapses under its own official one-epoch budget (three-seed $\rho = 0.096$), so the Geneformer result is not an artifact of overtraining, and scGPT-human at its official 15-epoch budget shows moderate average loss with architecture-specific variance, so the two architectures differ in collapse determinism rather than in collapse occurrence. Under official-convention budgets, the same-corpus comparison is thus budget-fair, and the earlier three-epoch arm remains the pre-registered budget-matched reference point rather than a claim about either model's prescribed usage.

The biological validation anchors these geometric measurements in published biology. Along the frozen and ordinal axes, the stemness program declines and the differentiation program rises monotonically, recovering the slow-cycling limbal stem-cell structure of Cotsarelis et al. [1]; on epidermis, every arm recovers the basal keratin program of Fuchs and Green [42]. Naive fine-tuning erases the graded biology in lockstep with the geometric collapse, so the erasure is a biological loss, not a mere rearrangement. Classification accuracy alone is therefore an insufficient criterion for evaluating adapted foundation models in differentiation biology: geometry-aware, layer-resolved metrics expose losses that aggregate accuracy conceals. Fine-tuning benefits are setting-dependent [15, 16]; adaptation can preserve gene–gene relationships when the context matches pretraining [49], so collapse reflects a mismatch between the supervisory signal and the geometry to be preserved, not an inevitable consequence

of adaptation. Encoding biological structure into the supervisory signal is one productive route: hierarchical cross-entropy losses over cell-ontology graphs improve out-of-distribution annotation [50], and our ordinal head extends the same principle from classification generalization to the preservation of ordinal geometry; EMP-FM's positive dual-task result [26] is consistent, because pseudotime regression provides a direct ordinal target for every cell, whereas centroid-based adjacency constrains only the already-overwritten readout layer. The contamination-aware paired benchmark released with this study is intended as a validation resource for such geometry-aware evaluation, in corneal biology and beyond: evaluations in this field remain predominantly within single datasets and settings, and leakage-prone splits can inflate apparent performance [51]. Geometry-faithful evaluation of this kind is a prerequisite for the reliable virtual-cell representations envisioned by the AI Virtual Cell initiative [52] and complements multi-capability benchmarks such as VCBench [53]. Reporting the gate-refused and unstable recipe arms as honest negatives is deliberate: the pre-specified stability gate was applied without post hoc tuning, and publishing such outcomes, paired with a constructive layer-resolved diagnostic, serves the community better than selective reporting [31, 32].

3.1. Limitations of the study

This study has five principal limitations. First, fine-tuned cross-architecture evidence spans two model families (Geneformer-V2-104M and scGPT-human) on a single tissue (normal human corneal epithelium); the external evidence base comprises a two-donor same-corpus holdout, a three-donor post-cutoff out-of-corpus test, and two cross-tissue epidermal datasets (GSE147424 and GSE202352), and whether the dissociation between gradient geometry and layer topology, the head-proximal overwrite pattern, and the domain-specific boundary generalize further across architectures, tissues, and larger donor panels remains to be tested. The scGPT-human fine-tuned arm carries additional structural limits: it uses a single test donor (Hu0216), so donor-level statistics degenerate and cell-level bootstrap confidence intervals are the primary uncertainty quantification; its ordinal axis has three tiers (the transit-amplifying class is unrepresentable under the locked gene filter), so cross-dataset ρ values are reported per dataset and not merged with the four-tier GSE153515 axis; and its conclusion is sensitive to the checkpoint convention, because the best-validation and final-epoch checkpoints differ in ρ (0.44 versus 0.50 mean, with a seed collapsing only in the final epoch), which we report as primary (best-val, the official scGPT convention) and sensitivity (final-epoch) rather than selecting post hoc. The frozen cross-architecture probes (scFoundation, UCE, and scGPT-human) each separated the four layers (probe macro-F1 0.79–0.95) but ranged from strong (scFoundation $\rho = 0.75$) to weak (scGPT-human $\rho = 0.22$) ordinal encoding on GSE153515, with all those external tests out-of-corpus for these models. The domain-specific collapse map reported here is therefore measured within Geneformer-V2-104M and scGPT-human on normal human corneal epithelium; transfer across further architectures and tissues is left to dedicated evaluation. In particular, a post-cutoff (out-of-corpus) single-cell epidermal test, which would separate the axis-encoding-strength effect from any residual corpus-exposure effect on the cross-tissue reversal, is left to future work. Second, the TAC class achieved F1 of 0.0 in every adapted run on GSE153515, reflecting extreme class rarity (171 of 14,775 external-test cells; 1.2%) rather than a method-specific failure, and the TAC class was absent altogether from GSE299152 (proliferation markers below 0.6% of cells);

although our sensitivity analysis confirms that excluding TAC raises macro-F1 uniformly across arms without changing the ranking, conclusions regarding TAC representations are constrained by sample size in both test sets. Third, the gradient-preserving recipe arms beyond the ordinal head were expanded to three seeds only for the two non-gated arms (AA-FT $\lambda = 1.0$, soft-Spearman $\lambda = 0.5$); flow was refused extension by its stability gate. The three-seed results confirm that AA-FT preserves the gradient only partially and that soft-Spearman is unstable across seeds, so neither is claimed as a reliable fix; these results bound the search space for gradient-preserving adaptation, directing future recipe design toward intermediate-layer constraints. Fourth, the pre-training audit places GSE153515 within the corpus at the DOI level (53,290 cells, organ = eye; the V1 audit found zero corneal datasets), places the cross-tissue epidermal control GSE147424 within the corpus as well (45,200 cells, organ = skin), finds 10.4% of training-atlas cells overlapping corpus DOIs, and detected no cell-level overlap in bounded Signal A sampling (0.54% of the corpus). The frozen model's same-corpus performance may therefore be inflated by pre-training exposure; our conclusions rest on the relative, fine-tuning-induced measurements, which are insensitive to any shared pre-training contribution, and the cross-tissue reversal is interpreted as an axis-encoding-strength effect (the frozen epidermal axis is weak, $\rho = 0.24$, despite in-corpus skin exposure) rather than a corpus-absence effect. Fifth, the pre-registration for the cross-architecture arm (G1/G0 gates, the van Zyl amendments, and the official-budget controls) was maintained as a project-level locked protocol amendment (PREREG §7–§9) rather than deposited in an external registry; all gate thresholds were fixed before the corresponding measurements, and the per-epoch convergence trajectories are archived on the training node (epoch-1–3 and epoch-15 points reproduced in Figure 6), but we did not publish a time-stamped registry entry, which limits independent verification of the pre-specified status of the amendments.

4. Methods

4.1. Data sources and preprocessing

Training atlas. Training and internal-test data were derived from the "Single cell atlas of the human ocular surface" collection (CELLxGENE Discover, collection identifier 0f7d022a-46c7-4e64-be4c-e34adbb78089). The data file used is the current dataset-version of CELLxGENE dataset 8ef06ea4-8d0f-4303-a652-7d4604402b1f (schema 7.1.0, published 2025-05-05), distributed as file 30127f02-d0b9-4b56-bf4c-e5b77e4989bd.h5ad (10.29 GB; MD5 39f098367af95bc3e1ee92528ceb251e; the file identifier is the dataset-version identifier at time of download, retained for traceability). The atlas is a pre-publication consortium product (CZI Cell Science / Human Cell Atlas Eye Network) without an associated journal publication or DOI, and is therefore cited as a CELLxGENE collection [9]. The collection aggregated ten constituent studies spanning corneal, limbal, scleral, and fetal ocular surface tissues [3–6, 8, 54, 55]. The aggregated AnnData object contained 120,000 cells with 20 curated cell-type labels and 31 author-assigned cell-type labels.

Cell-type labels were harmonized into a five-layer ordinal scheme reflecting the corneal epithelial differentiation continuum: LSC (including LPC and fetal epithelial progenitors; layer index 0), TAC (limbal epithelium; index 1), basal (Epi-PMC; index 2), wing (Epi-TDC/PMC; index 3), and squamous (Epi-TDC/unknown epithelium; index 4). Non-epithelial populations (immune, endothelial, stromal, conjunctival, melanocyte) were assigned layer index -1

(NON_LAYER) and excluded from ordinal analyses. For the four-class classification task evaluated on external data, the scheme was collapsed to LSC, TAC, basal, and differentiated (wing and squamous merged).

External test datasets. The study paired two external test sets with opposite pre-training-corpus status (Methods, Provenance and contamination audit). GSE153515 (GEO accession), a 10x Genomics scRNA-seq dataset of normal human limbal basal epithelium [3, 6], contained 17,587 cells from two donors (GSM4646295, GSM4646296) with an initial gene matrix of 32,738 features and is the same-corpus holdout. GSE299152 [36], a single-cell map of the ocular surface glycocalyx in adult human and mouse (human cells used here), contained 26,628 human cells from three donors with an initial gene matrix of 36,601 features and is the post-cutoff, out-of-corpus test. Disjointness from the training atlas was verified programmatically via the CELLxGENE Discover API: no overlapping GEO accession, donor identifier, or dataset hash was found between either external test and any of the ten constituent training studies; disjointness from the pre-training corpus is addressed in the contamination audit.

4.2. Fair donor-disjoint splitting protocol

To prevent donor-level information leakage between training and evaluation, a strict donor-disjoint split was applied to the training atlas prior to any model fitting. Donors were partitioned using a deterministic random assignment (seed = 42), yielding 49 donors (99,606 cells) for training and 25 donors (20,394 cells) for internal testing, with zero donor overlap between partitions. The split protocol ensured representation of all 20 curated labels and all 31 author-assigned labels in both partitions. The split manifest, including per-partition cell counts, donor lists, label distributions, and SHA-256 checksums of the resulting train and test AnnData files, was locked and archived (artifacts/paper1_fair_split/manifest.json). No cells were reassigned after manifest generation.

4.3. External test annotation pipeline

Because GSE153515 was deposited with an empty observation metadata table, a fully reproducible de novo annotation pipeline was implemented (scripts/protocol_15_annotate_gse153515.py): (1) donor assignment from the 10x barcode suffix convention; (2) marker scoring on the full 32,738-gene matrix using eight curated marker gene sets spanning LSC, TAC, basal, differentiated, conjunctival, melanocyte, immune, and stromal identities; (3) quality-control filtering by minimum gene count and maximum mitochondrial fraction (15,827 of 17,587 cells retained, 89.9%, spanning 10,605 genes); (4) integration and clustering: highly variable gene selection, log-normalization, scaling, PCA, Harmony batch correction (n_pcs = 30) [56], and Leiden clustering (resolution 0.6) yielding 16 clusters; (5) DEG-based cell-type assignment via Wilcoxon rank-sum test with an enrichment score $(n_{\text{in top200}}/n_{\text{total}}) \times (1 - \text{mean_rank}/200)$ against the top 200 cluster-specific DEGs, with clusters below an epithelial enrichment threshold of 0.3 classified as NON_LAYER; and (6) harmonization to the four-class scheme (basal 6,698; differentiated 4,483; LSC 3,423; TAC 171 cells) with non-epithelial cells excluded. All annotation parameters were locked in a configuration file (config/external_test_gse153515.yml) before any model evaluation. The second external test, GSE299152, was annotated with a mirrored pipeline (scripts/protocol_15b_annotate_gse299152.py; config/external_test_gse299152.yml) with two recorded deviations: Endothelial

and Stromal marker sets were added to the NON_LAYER gate, and Leiden resolution was auto-tuned to 0.7 (15 clusters). After QC (23,970 of 26,628 cells retained, 90.0%; 11,274 genes), harmonization yielded LSC 15,921, basal 1,099, and differentiated 4,354 cells, with an empty TAC class (proliferation markers below 0.6% of cells) and 2,596 non-epithelial cells excluded; low-confidence clusters (melanocyte enrichment 0.199 below the 0.3 gate; MT-high and stress clusters with near-zero enrichment) were default-assigned to LSC/LPC and flagged in the annotation report.

4.4. Model architectures

Four model configurations were evaluated under an identical experimental protocol (same preprocessing, data split, classification head, training epochs, and random seeds); two additional ordinal-head arms (Arms 5–6) and five gradient-preserving objective arms (Arms 7–11) shared the same protocol and budget:

Arm 1—HVG/PCA + linear classifier (transparent baseline). Highly variable genes were selected from the training partition; expression values were log-normalized and scaled, PCA was applied, and a linear classifier was trained on the resulting principal components.

Arm 2—Frozen Geneformer + linear probe. Geneformer-V2-104M (BertForMaskedLM architecture; hidden dimension 768, 12 layers, 12 attention heads, vocabulary 20,275) was loaded from the Hugging Face model hub (snapshot commit 04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5; model weights SHA-256 fff5cba2…), pretrained on Genecorpus-104M (∼104 million cells from 2,903 datasets) [11, 12]. All Transformer parameters were frozen; only a linear classification head appended to the [CLS] token representation was trained.

Arm 3—Geneformer naive fine-tuning ($\lambda = 0$). The full model was fine-tuned end-to-end with a standard cross-entropy classification loss over the 31 author-assigned cell-type classes (no continuity regularization), serving as the fair adaptation comparator; predictions were merged to the four epithelial differentiation classes for external evaluation.

Arm 4—Geneformer + CAFT ($\lambda = 0.5$). The full model was fine-tuned with the CAFT composite loss (Methods, CAFT loss formulation) at a single fixed regularization strength, testing CAFT as a candidate safeguard for ordered differentiation geometry.

Arm 5—Geneformer + ordinal-naive head ($\lambda = 0$). The full model was fine-tuned end-to-end with the CORN ordinal loss (Methods, Ordinal head formulation) and no continuity regularization.

Arm 6—Geneformer + ordinal-CAFT head ($\lambda = 0.5$). The full model was fine-tuned with the CORN ordinal loss plus the CAFT continuity term, wired through a registered Trainer callback so that the continuity objective was genuinely executed (in contrast to the archived CAFT arm).

Arms 7–11—axis-anchored, soft-Spearman, and flow objectives. Three further gradient-preserving objectives were evaluated with a four-class cross-entropy head on the revised centroid-PCA differentiation axis (Methods, Axis definition revision): Arm 7 (AA-FT, $\lambda = 0.5$) and Arm 8 (AA-FT, $\lambda = 1.0$) augment the classification loss with a projection-constraint term pulling embeddings onto the fixed differentiation axis [57]; Arm 9 (soft-Spearman, $\lambda = 0.5$) and Arm 10 (soft-Spearman, $\lambda = 1.0$) add a differentiable Spearman surrogate (soft rank via sigmoid pairwise comparisons with a diagonal correction, ABAG-Rank

formulation [58], with a margin surrogate [59]; a differentiable sorting-and-ranking method [60]); and Arm 11 (flow, $\lambda = 0.5$) adds a flow-consistency objective constraining embeddings toward ordered transitions between adjacent layers. These arms used the same preprocessing, data split, and training budget as Arms 3–6.

Arms 3–6 were each trained with three independent random seeds (42, 123, 456) for three epochs. Arms 7–11 were trained with seed 42 only: a three-seed expansion was gated on the seed-42 evaluation reaching a pre-specified differentiation-gradient range (0.5–0.7) on the same-corpus holdout, which no arm reached (Results). Training used a per-device batch size of 8 with gradient accumulation of 4 (effective batch size 32), bfloat16 mixed precision, gradient checkpointing, linear learning-rate warmup over 500 steps, weight decay 0.001, and learning rate 5×10^{-5}.

4.5. CAFT loss formulation

The CAFT objective augmented standard cross-entropy classification with a continuity regularization term:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \cdot \mathcal{L}_{\text{cont}}, \tag{1}$$

where $\mathcal{L}_{\text{cls}}$ is the cross-entropy classification loss over the 31 author-assigned cell-type classes of the training atlas (merged to the four epithelial differentiation classes for external evaluation) and $\mathcal{L}_{\text{cont}}$ penalizes deviation from expected adjacency structure. For each training cell x_i with assigned differentiation layer $\ell_i \in \{0, 1, 2, 3, 4\}$, the continuity loss was

$$\mathcal{L}_{\text{cont}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{\ell' \in \mathcal{A}(\ell_i)} \cos\Big(\mathbf{z}_i,\ \text{sg}(\mathbf{c}_{\ell'})\Big), \tag{2}$$

where $\mathbf{z}_i$ is the ℓ_2-normalized [CLS] embedding of cell x_i, $\mathcal{A}(\ell_i)$ denotes the adjacent layers ($\ell_i \pm 1$, clipped to $[0, 4]$), $\mathbf{c}_{\ell'}$ is the centroid of training embeddings assigned to layer ℓ', and $\text{sg}(\cdot)$ is the stop-gradient operator. Layer centroids were precomputed at the start of each epoch from detached embeddings and held fixed throughout that epoch. NON_LAYER cells ($\ell = -1$) were excluded from the continuity term but contributed to $\mathcal{L}_{\text{cls}}$.

4.6. Ordinal head formulation

To test whether the collapse of the differentiation gradient is caused by treating the four ordered layers as unordered softmax classes, Arms 5–6 replaced the softmax classification head with a cumulative ordinal head following the CORN formulation [39]. For $K = 4$ ordinal classes, the head emits $K - 1 = 3$ binary threshold logits, where threshold k predicts whether the cell has reached ordinal stage k or beyond ($P(y \geq k)$); the loss is the sum of the three binary cross-entropies, masked to epithelial cells:

$$\mathcal{L}_{\text{ord}} = \sum_{k=1}^{K-1} \text{BCE}\Big(\sigma(f_k(\mathbf{z})),\ [y \geq k]\Big), \tag{3}$$

where $f_k(\mathbf{z})$ is the k-th threshold logit on the [CLS] representation $\mathbf{z}$, σ is the sigmoid, and $[y \geq k]$ is the binary target indicating that the cell's ordinal layer is at least k. At inference the predicted class is the number of thresholds with $\sigma(f_k(\mathbf{z})) \geq 0.5$. The CORN head makes no equidistance assumption on the thresholds and yields class probabilities directly for

calibration ($P(y = k) = p_k - p_{k+1}$, with $p_0 = 1$, $p_K = 0$). The ordinal-CAFT arm (Arm 6) added the CAFT continuity term ($\mathcal{L} = \mathcal{L}_{\text{ord}} + \lambda \cdot \mathcal{L}_{\text{cont}}$), with $\mathcal{L}_{\text{cont}}$ wired through a registered `TrainerCallback` so that the continuity objective was genuinely executed at each epoch start, in contrast to the archived CAFT arm, where the equivalent hook was never dispatched.

4.7. Evaluation metrics

All metrics were computed on the locked external test sets using the Tier-B evaluation pipeline (corneal_fm/evaluation.py).

Primary endpoint. Donor-level macro-averaged F1 (macro-F1) on the four-class external test set, with 95% confidence intervals estimated by nonparametric bootstrap resampling of donors (n_boot = 1,000; seed = 42). Per-class precision, recall, F1, and support were reported for each arm and seed.

Secondary endpoints. (i) Differentiation-gradient rank correlation (Spearman ρ): an ordinal-structure metric computed solely from the layer-centroid geometry of the learned embeddings, without marker-gene expression. For each of the four layer classes (NON_LAYER excluded), the centroid of the cell embeddings was computed; the principal differentiation axis was defined as the first singular vector of the centered layer centroids, oriented so that pseudotime increases from LSC to differentiated; each cell was assigned a pseudotime equal to the projection of its embedding onto this axis; and ρ was the Spearman correlation between per-cell pseudotime and the a priori layer ordinal. (ii) Layer Adjacency Accuracy (LAA): the fraction of nearest-neighbor pairs in the learned embedding space belonging to adjacent or identical differentiation layers. (iii) Expected Calibration Error (ECE): the weighted mean absolute difference between predicted confidence and observed accuracy across probability bins. (iv) Internal silhouette coefficient and gap statistic were computed as diagnostic-only measures and were not used for model selection or promotion decisions.

Layer-resolved gradient measurement. At inference, all thirteen hidden states (embedding output L0 and layers L1–L12) were collected for each of the 14,775 external-test epithelial cells, and the differentiation-gradient ρ was recomputed at each depth from the position-0 [CLS] representation (scripts/measure_layerwise_rho.py). A sanity gate required the L12 ρ of every model to reproduce the corresponding single-value measurement within 10^{-6} (all seven models passed bit-exactly). At L0 the position-0 [CLS] representation is constant across cells, so ρ is undefined (NaN) and was excluded (expected behavior). For the randomized-head ablation, the trained classification head was replaced with a freshly initialized, untrained head and hidden states were collected without further training; the maximum hidden-state difference from the intact model was 0 for both the fine-tuned and frozen bodies. Cell-level bootstrap 95% confidence intervals at L12 were computed with n_boot = 1,000 resamples (seed = 42; scripts/bootstrap_layerwise_rho.py).

Trajectory inference. Diffusion pseudotime (scanpy DPT) was computed on PCA-reduced external embeddings (n_pcs = 30, n_neighbors = 30, n_dcs = 10), rooted at the LSC class, and correlated (Spearman) with the reference layer-ordering pseudotime for each arm (scripts/analyze_trajectory_continuity.py).

TAC sensitivity analysis. Because the TAC class is extremely rare (171/14,775 cells; 1.2%), donor-level macro-F1 was recomputed excluding TAC (three classes), with the same bootstrap procedure, to assess whether the rare class drives the primary endpoint (scripts/analyze_tac_sensitivity.py).

CNS-anchored marker-program validation. To test whether the measured axis carries published biology, per-cell marker-program scores were computed as the mean of z-scored log1p expression over literature-anchored gene programs: stemness (KRT15, KRT19, TP63), corneal differentiation (KRT12, KRT3), and cycling (PCNA, MCM2, MKI67, TOP2A) on GSE153515; basal (KRT5, KRT14, COL17A1) and differentiation (KRT1, KRT10) keratins and cycling (MKI67, PCNA) on both epidermal datasets (GSE147424 and GSE202352). Expression was read from the raw GSE153515 matrix (barcode-aligned to the 14,775 epithelial cells; the QC-filtered matrix drops MKI67) and from the processed epidermis objects; GSE202352 (Wiedemann et al. [43], Cell Reports 2023) was assembled from its 10x filtered feature-barcode matrices (11 of 12 samples; one sample's matrix is corrupt at the GEO source and was excluded, covering all four donors and all three body sites) with the same marker-axis quartile binning as GSE147424 (scripts/process_gse202352.py; scripts/run_gse202352_eval.py). Each program score was correlated (Spearman) with the arm's per-cell axis pseudotime (axis definition, above); uncertainty was quantified by nonparametric bootstrap (500 resamples; seed 42) and permutation nulls (500 label-permutation replicates) standardized as z-scores (scripts/bio_validation.py). Circularity disclosure: KRT15, KRT19, KRT12, KRT3, and MKI67 appear in the locked annotation marker sets, so class-level enrichment of the corresponding programs is partially annotation-derived; the axis measurement and the program correlation are nevertheless distinct quantities, and the independent markers NFIB, CDH3, PCNA, and MCM2 (absent from all annotation sets) corroborate the recovery on the cornea. On epidermis, the basal-keratin and cycling programs are absent from the marker-score definition used to construct the ordinal layer labels.

4.8. Axis definition revision (protocol deviation)

The axis-anchored and soft-Spearman arms (Arms 7–10) were initially trained against a marker-score differentiation axis. An evaluation of seed 42 exposed a training-axis versus evaluation-metric direction mismatch: the marker-score axis and the per-layer-centroid-PCA axis used by the evaluation metric (Secondary endpoints) aligned with cosine 0.470. On 2026-08-10 the axis definition was revised (protocol deviation, user-approved; changelog 2026-08-10-paper1-axis-refit-option-a) to a per-layer-centroid-PCA axis computed over all epithelial training cells, bit-identical to the evaluation metric's axis (corneal_fm/axis_ref.py, mode = "centroid_pca"; cache schema v2); the marker-score axis is retained as a reference mode. Arms 7–11 and the cross-tissue evaluation used the revised axis.

4.9. Cross-architecture frozen probes (scGPT-human, scFoundation, UCE)

To probe whether the ordinal-geometry measurements generalize across single-cell foundation models without incurring a second fine-tuning protocol, the frozen embeddings of scGPT-human (perturblab/scgpt-human checkpoint; 12-layer Transformer, 512-dimensional hidden state, continuous log1p input, CLS cell embedding, 51.86M parameters; scgpt 0.2.5, git commit cebd6fae) were evaluated on both external test sets. Cells were tokenized with the model's 60,697-gene vocabulary (append_cls = True, maximum sequence length 1,200 with the CLS token kept first) and embedded at the CLS position of the final layer; the differentiation-gradient ρ was computed with the same marker-gradient metric as all other arms, and class separability was measured with a four-class logistic-regression probe (cell-level 80/20 split, five repeats, mean macro-F1). Gene-symbol mapping for GSE153515 was verified

against HGNC: 10,379 of 10,605 ENSEMBL identifiers resolved authoritatively, of which 32 of the 9,175 vocabulary-matched genes (0.35%) carry post-download HGNC renaming (none core corneal markers), with no effect on the input.

Frozen scFoundation (01B-resolution checkpoint; genbio-ai/scFoundation Hugging Face mirror, checkpoint sha256 9f40bf32…; xTrimoGene encoder–decoder, 119.25M parameters [61]) was embedded with the official inference pipeline (get_embedding.py; input_type singlecell, output_type cell, pool_type all, tgthighres a5, pre_normalized F, version ce), yielding per-cell embeddings of dimension 3,072 (concatenation of last-layer, second-to-last, max-pooled, and mean-pooled hidden states over the 19,264-gene vocabulary). Raw counts were aligned to the model's 19,264 HGNC-symbol vocabulary (9,162 of 9,175 input genes matched for GSE153515; 10,296 of 11,274 for GSE299152; unmatched genes zero-filled). scFoundation's pretraining corpus (over 50 million human cells) is not publicly enumerated, so corpus membership of the two test sets could not be verified and is reported as inferred out-of-corpus.

Frozen UCE (Universal Cell Embedding; 33-layer checkpoint 33l_8ep_1024t_1280.torch, Figshare 10.6084/m9.figshare.24320806, checkpoint sha256 3f458726…; [18]) cell embeddings were computed with the official inference script (eval_single_anndata.py; species human, sample_size 1,024 expression-weighted genes sampled per cell with replacement, pad_length 1,536, CLS-token hidden state of dimension 1,280, L2-normalized). Because UCE cell representations depend on stochastic gene sampling, each dataset was embedded three times (seeds 42, 123, 456) and the differentiation-gradient ρ and probe macro-F1 were averaged across repeats (per-repeat values in Table S8). UCE was pretrained on the CELLxGENE census 2023-04-25 release; GSE153515 is absent from the closest queryable census release (2023-05-15) and GSE299152 postdates the corpus, so both test sets are reported as out-of-corpus (inference).

4.10. Cross-architecture fine-tuning protocols (van Zyl same-corpus arm and official-budget controls)

Epoch-convention correction. The fine-tuning literature specifies official training budgets that differ by model: Geneformer's official protocol trains for a single epoch to avoid overfitting [11], whereas the official scGPT protocol trains for 15 epochs and selects the checkpoint by best validation score [13]. The main protocol of this study (three epochs, shared across arms for budget-matched comparison) is therefore an over-training regime for Geneformer and an under-training regime for scGPT; the cross-architecture arm below aligns each model to its official convention (a pre-registered amendment; PREREG §9), so that the comparison reflects the models' own prescribed usage rather than a shared ad hoc budget.

van Zyl same-corpus arm. scGPT-human (perturblab/scgpt-human; 12-layer Transformer, 512-dimensional hidden state, continuous log1p input, CLS cell embedding, 51.86M parameters; scgpt 0.2.5, git commit cebd6fae) was pretrained on the CELLxGENE census 2023-05-15 release; its only corneal dataset is the van Zyl et al. limbal and central cornea atlas [40]. The same-corpus training set comprised 62,640 epithelial cells (leiden cluster 0 excluded under a pre-registered amendment after forensic adjudication of 13,584 ciliary-muscle cells; three-tier ordinal axis LSC→basal→differentiated, the transit-amplifying class being unrepresentable because all six cycling markers were removed by the locked gene filter; PREREG §7). The locked donor split assigned the largest-epithelial donor (Hu0216; 20,158 cells) as test

donor and the remaining four donors (42,482 cells) for training (PREREG §8). Two gates were locked before training: G1 (frozen-axis existence, $\rho \geq 0.5$ and label-shuffle $z \geq 3$) and G0 (training adequacy; revised pre-registration: $\mathrm{F1}_{\mathrm{ce}} \geq 0.55$ and $\geq \mathrm{F1}_{\mathrm{frz}} - 0.10$ with $\mathrm{F1}_{\mathrm{frz}} = 0.9494$). Frozen embeddings, ρ/z computation, and the frozen probe followed the same metric implementation as the main protocol (Evaluation metrics), restricted to the three-tier axis.

Convergence protocol (15 epochs). scGPT-human was fine-tuned with naive cross-entropy over the 30 author cell-type classes present in the training cells, under the locked hyperparameters of the main protocol (per-device batch 8 with gradient accumulation 4, effective batch 32; learning rate 5×10^{-5}, weight decay 0.001, linear warmup over 500 steps; bfloat16; maximum sequence length 4,253). A validation partition of 4,247 cells (10%) was carved from the 42,482 training cells by stratified sampling over donor $\times$ layer class (seed 42; split manifest locked with SHA-256 before training; the test donor never enters validation or training). Each epoch saved a checkpoint and recorded validation macro-F1, validation ρ (L2-normed and raw), and a label-shuffle z (n = 100) in a per-epoch trajectory; the best-validation checkpoint was selected by validation macro-F1 (the official scGPT best-validation-score convention), with the final-epoch checkpoint retained as a sensitivity item. Both checkpoints were evaluated on the locked test donor with the same metric suite as the main protocol (cell-level bootstrap for the confidence intervals, because a single test donor degenerates donor-level statistics; PREREG §8).

Geneformer official-budget control. Geneformer naive cross-entropy fine-tuning was repeated under the official one-epoch budget (all other hyperparameters identical to the main protocol), on the same donor-disjoint training partition and locked GSE153515 external test as the main arm, with three seeds.

4.11. CAFT promotion criteria

CAFT (Arm 4) was promoted over naive fine-tuning (Arm 3) only if all four criteria were satisfied simultaneously: (1) external continuity improvement (ρ or LAA) relative to Arm 3; (2) non-inferior classification (bootstrap 95% CI overlap or non-inferiority within a prespecified margin); (3) seed and donor consistency of the improvement direction; and (4) survival of the locked provenance audit and donor- and study-level disjointness verification. Failure of any single criterion retained Arm 3 as the preferred strategy, with CAFT reported as an unsuccessful candidate; no post hoc tuning or selective reporting was permitted.

4.12. Provenance and contamination audit

All computational artifacts were version-locked prior to analysis: model weights (retrieved 2026-07-23), training and test data files (downloaded 2026-07-22), and SHA-256 checksums for model weights and train/test AnnData were recorded, and the fair-split manifest was generated before any training. Because Geneformer was pretrained on Genecorpus-104M, we assessed whether corneal or ocular surface datasets from our training atlas might have been included in the pretraining corpus. For Genecorpus-30M (V1; 561 datasets with a public manifest), all 43 eye-related datasets were identified; all were retinal or retinal pigment epithelium datasets, with zero corneal datasets, and all eight GEO accessions corresponding to our training studies returned no matches. For Genecorpus-104M (V2; 2,903 datasets), the official corpus source table (Supplementary table MOESM3, sheet `genecorpus_104m`; 563 unique DOIs) was

downloaded from the Nature Computational Science supplementary materials and used for a DOI-level audit. GSE153515's primary publication DOI (10.1016/j.jtos.2020.12.004; Li et al., The Ocular Surface 2021) resolves to a row in the corpus table (53,290 cells, organ = eye); because all corpus datasets predate assembly and GSE153515 was public from 2021-01-14, no temporal exemption applies, and GSE153515 is reported as same-corpus (least conservative). GSE299152 [36] (GEO public 2025-11-18) postdates corpus assembly and its DOI is absent from the table, making it the post-cutoff, out-of-corpus (most conservative) test. The cross-tissue control GSE147424 [37] (He et al., J Allergy Clin Immunol 2020) likewise resolves to a row in the corpus table (organ = skin, 45,200 cells), so the epidermal test is within the pre-training corpus; its frozen differentiation axis is nevertheless weak ($\rho = 0.24$ versus 0.81 for the same-corpus corneal test), showing that corpus membership and axis encoding strength are decoupled (Results, three-domain boundary map). The second cross-tissue dataset, GSE202352 [43] (Wiedemann et al., Cell Reports 2023; organ = skin, 15,243 cells), is also within the corpus; both epidermal tests are therefore reported as in-corpus cross-tissue controls, and the cross-tissue reversal is interpreted as an axis-encoding effect rather than a corpus-absence effect. For the cross-architecture probe, scGPT-human was pretrained on the CELLxGENE census 2023-05-15 release (33,758,887 unique normal human cells; [13] Data Availability). GSE153515 is absent from both the CELLxGENE curation catalog and the census 2023-05-15 dataset manifest (verified via the CZ curation API and the census S3 h5ads directory), and GSE299152 (GEO released 2025-11-18) postdates the corpus; both external tests are therefore out-of-corpus for scGPT. A secondary finding: two training-atlas studies (lako_GSE155683, 9,972 cells; li_GSE147979, 423 cells) resolve to corpus DOIs (10,395 of 99,606 cells, 10.4%), disclosed for completeness; this does not affect the external-test conclusions, which rest on donor- and study-level disjointness from the training atlas. As a corroborating cell-level check in the spirit of scContam [62], bounded Signal A fingerprint sampling (0.54% of the corpus, 491,227 fingerprints; vectorized MinHash m = 128, b = 16, r = 8, banded LSH, exact Jaccard confirmation) found zero hits for the 15,827 GSE153515 cells (max Jaccard 0.0), with 100% positive-control recall and zero negative-control false positives; because full-corpus coverage (approximately 600 GB) was not attained, this run validates the pipeline but does not constitute exclusion evidence. The DOI-level audit is the primary evidence.

CAFT implementation audit. A post hoc audit established that the continuity term $\mathcal{L}_{\text{cont}}$ never executed: in Transformers 4.57 the `on_epoch_begin` hook is dispatched only to registered `TrainerCallback` objects, and `CAFTTrainer.on_epoch_begin` was a plain method that no callback ever invoked, so `layer_centroids` was never populated and every archived run trained pure cross-entropy (archived scripts byte-identical; training histories contain only the `loss` key). The $\lambda = 0.5$ checkpoints' higher final-layer ρ ($+0.210$ versus -0.008) is therefore run-to-run nondeterminism, not restoration (audit forensics and corrected ECE values in artifacts/paper1_pipeline_audit/). A correctly wired continuity objective was subsequently tested inside the ordinal-CAFT arm, where it executed genuinely but conferred no measurable benefit beyond the ordinal head (Methods, Ordinal head formulation; Results, ordinal head).

4.13. Statistical analysis plan

All statistical analyses were prespecified in the locked evaluation configuration prior to model training. The primary comparison was between Arm 3 and Arm 4, with Arms 1 and 2 as reference baselines. All primary same-corpus claims were submitted to a single Benjamini–Hochberg FDR family of 26 comparisons (q threshold 0.05), spanning, per arm, gradient-positivity ($\rho > 0$), layer-adjacency-above-chance (LAA > chance), and classification-parity (F1 = naive F1) tests across the frozen, naive, CAFT, ordinal, AA-FT, soft-Spearman, and flow arms (scripts/fdr_nulls_analysis.py; artifacts/paper1_fdr_nulls/bh_fdr_table.json). Permutation nulls were constructed by label shuffling (500 replicates) and feature shuffling (300 replicates) of the differentiation-gradient measurement, and observed ρ values were standardized against the null mean and SD (z-scores; artifacts/paper1_fdr_nulls/null_controls.json). Uncertainty was quantified by nonparametric bootstrap resampling at the donor level (n_boot = 1,000; seed = 42). Model stability was assessed across three independent training seeds, reported as mean ± range. All hypothesis tests were two-sided; $p < 0.05$ was considered significant. No interim analyses or adaptive stopping rules were employed.

4.14. Software and reproducibility

All analyses were performed in Python 3.12, using Scanpy [63], harmonypy [56, 64], Hugging Face Transformers [65], PyTorch [66], and scikit-learn [67]. The CAFT training pipeline was implemented as a custom extension of the Hugging Face Trainer class (corneal_fm/caft.py); evaluation used the Tier-B module (corneal_fm/evaluation.py); external annotation used the locked pipeline scripts and configuration files. Gradient-preserving objective training used corneal_fm/axis_ref.py and scripts/method_arm_gf.py; the three-domain boundary evaluation used scripts/run_method_arms_eval.py, scripts/epidermis_tpoac_eval.py, and scripts/summarize_tpoac_boundary.py; FDR and permutation-null control used scripts/fdr_nulls_analysis.py; the CNS-anchored marker-program validation used scripts/bio_validation.py and scripts/extract_ordinal_embeddings.py.

Resource availability

Lead contact. Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Qi Dai (dq@mail.eye.ac.cn).

Materials availability. This study did not generate new unique reagents or biological materials; all data are public (below).

Data and code availability.

- **Training data.** CELLxGENE Discover collection 0f7d022a-46c7-4e64-be4c-e34adbb78089 (Single cell atlas of the human ocular surface; Human Ocular Surface Atlas), current dataset-version of dataset 8ef06ea4-8d0f-4303-a652-7d4604402b1f, distributed as file 30127f02-d0b9-4b56-bf4c-e5b77e4989bd.h5ad; MD5 39f098367af95bc3e1ee92528ceb251e.

- **External test data.** GEO GSE153515 [3]; 17,587 cells from two donors (GSM4646295, GSM4646296). GEO GSE299152 [36]; 21,374 epithelial cells from three donors after QC. Cross-tissue: GEO GSE147424 [37].

- **Cross-architecture probes.** scGPT-human checkpoint (perturblab/scgpt-human; scgpt 0.2.5, git commit cebd6fae); scFoundation 01B-resolution checkpoint (genbio-ai/scFoundation Hugging Face mirror); UCE 33-layer checkpoint (Figshare 10.6084/m9.figshare.24320806); frozen-probe diagnostics in artifacts/scgpt_repro/frozen_diag/, artifacts/scfoundation_repro/frozen_diag/, and artifacts/uce_repro/frozen_diag/.

- **Model.** Geneformer-V2-104M, Hugging Face model hub (ctheodoris/Geneformer; snapshot commit 04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5).

- **Code.** Analysis code is available at `https://github.com/Xu-Kunk/Cornea-Ai` and will be made publicly accessible upon acceptance.

- **Evaluation scripts.** scripts/run_tier_b_eval.py; scripts/run_tier_b_baselines.py; scripts/measure_layerwise_rho.py; scripts/bootstrap_layerwise_rho.py; scripts/analyze_trajectory_continuity.py; scripts/analyze_tac_sensitivity.py; scripts/protocol_15b_annotate_gse299152.py; scripts/method_arm_gf.py; scripts/run_method_arms_eval.py; scripts/epidermis_tpoac_eval.py; scripts/process_gse202352.py; scripts/run_gse202352_eval.py; scripts/summarize_tpoac_boundary.py; scripts/fdr_nulls_analysis.py; scripts/bio_validation.py; scripts/extract_ordinal_embeddings.py.

- **Results.** artifacts/paper1_tier_b_eval/*.json (six fine-tuned-arm and two baseline-arm result files); artifacts/paper1_tier_b_eval_gse299152/*.json (out-of-corpus results); artifacts/paper1_ordinal_eval/*.json (ordinal arms); artifacts/paper1_epidermis_eval/epidermis_rho_results.json (cross-tissue, GSE147424); artifacts/paper1_epidermis_eval_gse202352/gse202352_rho_results.json (cross-tissue, GSE202352); artifacts/paper1_bio_validation/bio_validation_results.json (CNS-anchored marker-program validation); artifacts/paper1_fdr_nulls/ (bh_fdr_table.json, null_controls.json, contamination_audit_protocol.md).

- **Reproducibility.** All experiments are reproducible with fixed random seeds (42, 123, 456), locked configuration files (config/external_test_gse153515.yml, config/external_test_gse299152.yml), and the SHA-256-locked fair-split manifest.

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